THE STORY BEHIND THE EXPERIMENT

The curled torus burps

A small push, a large release,
and the question of what survives

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Everything you have ever seen happened in space with three large directions you can move in. We know its rules well. What we do not know is how it came to be.

This paper is about a hypothetical change: space emerging sharply from something else. We want to understand that change, and how the two phases of being are related. We engage a toy model of dots and links and ask three questions: what does it take to start such a change, what comes out when it happens, and what is left behind.

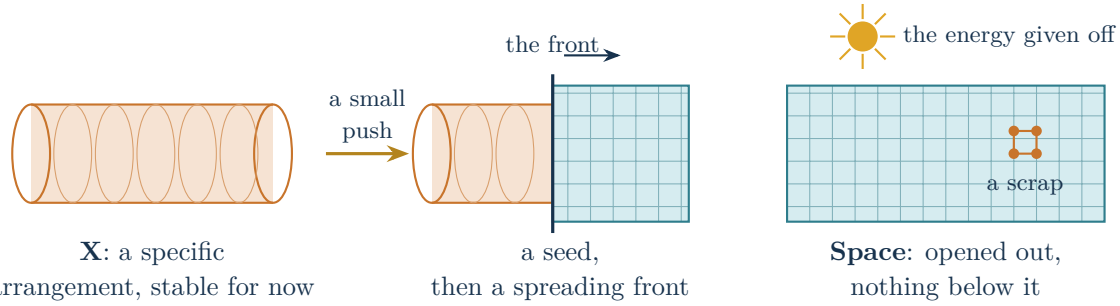


Figure 1. The idea in one picture. SKETCH, NOT DATA What came before space, called X here (left), waits until a small push starts it; the new arrangement spreads behind a front (middle); when it is done, flat space holds the energy that came out plus a small scrap of the old arrangement (right). Everything in this paper is a test of some part of this proposed picture inside one toy model. A single front and the large opened space are hypotheses, not established outcomes of these runs. Total energy includes the reservoir. Chapter 3 says why X is drawn as a tube; Chapter 6 shows how its measured endpoint differs.

Who is saying what

The project keeps four kinds of statement apart, and this edition tags them where it matters, so that a sentence is never quietly promoted from one kind to another. One more tag marks other people's published work.

- EXACT Arithmetic, or a complete listing of every possibility. True of the toy whatever a run does.
- MEASURED A simulation whose rules and predictions were written down before it ran, with its verdict. MEASURED, EXPLORATORY means a run that was not formally pre-registered: its prediction was only written into its configuration, or it is a closer look at runs already read.
- PUBLISHED Other researchers' published results.

- OUR READING, UNVERIFIED A reading of a result, offered by the author and her AI assistants and checked by no physicist. THE AUTHOR'S CLAIM The author's hypothesis about reality, stated as hers.

Numbers in square brackets point to the sources listed at the end.

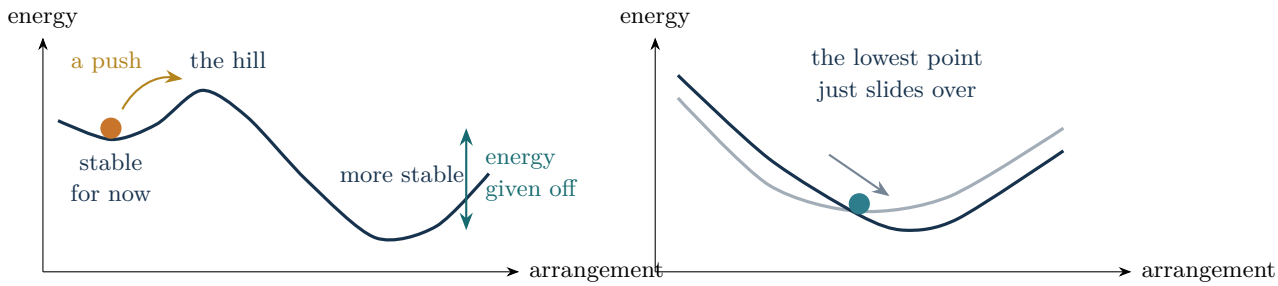
1 THE PLACE WE START

We live in space, and we cannot rerun its birth

PUBLISHED On the largest scales, observations are consistent with zero spatial curvature within the cosmological analysis used to interpret them [10]. That does not mean space is flat everywhere: gravity curves spacetime around stars and planets. It has three large directions. Its rules are known: general relativity for gravity and the shape of space, quantum mechanics for everything small.

What is not known is how such a thing comes to exist. Candidate theories say that space is not the stage on which everything happens but a *phase* of something deeper, the way ice is a phase of water. If that is right, there was a change, and changes come in two kinds.

PUBLISHED Cool water, and nothing much happens until it reaches zero degrees. Then it freezes, giving off heat as it goes, with ice and water side by side for a while. Very clean water can even stay liquid below zero, stuck behind a hill it cannot climb on its own, until a jolt sets it off. Heat a fridge magnet instead and its pull fades away: no hill, no burst of heat, no moment when half of it has changed. Physicists call the first kind **first order** and the second **continuous**. Plainer words: sharp and smooth.



Activated escape: a hill and a release **A smoothly changing preference**

Figure 2. A landscape analogy. SKETCH, NOT DATA The left picture illustrates activated escape from a local energy minimum; the right illustrates a smoothly changing preferred arrangement. These energy sketches are not a test of thermodynamic order. At nonzero temperature, phase stability depends on free energy, which includes the number of available arrangements as well as their energy.

How can you test which kind you have? PUBLISHED Near coexistence, a well-sampled energy histogram can show two humps: the system spends time in each phase. But one hump in a small system does not prove a continuous transition, and two humps alone do not establish a first-order transition. A small system can show two humps that merge as it grows. First order means that the humps stay apart and the valley between them deepens as the system gets larger, so the test is what happens with size. Here that equilibrium test remained inconclusive, and this paper does not rest on it. Neither does the author's idea. What she asks of the toy, and what is measured here, is plainer: does a prepared tube give way with a jump in energy at the sizes run, and how large is the jump?

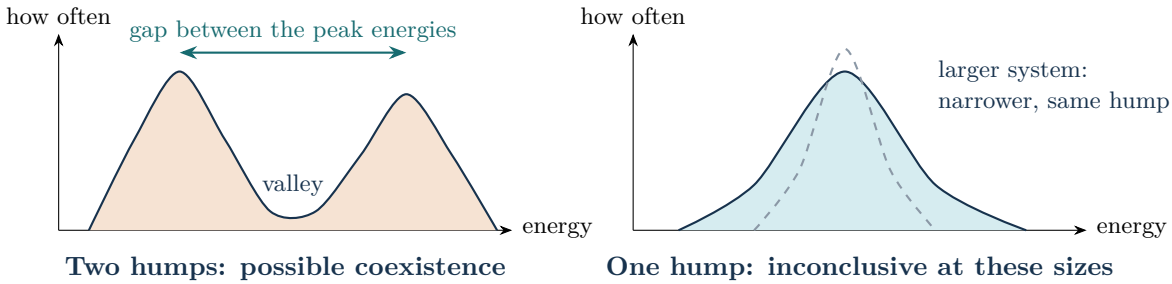


Figure 3. What an equilibrated run can measure. SKETCH, NOT DATA An energy histogram depends both on the cost of arrangements and on how many have each energy. Near coexistence, separated peaks can estimate a finite-size energy gap. Classifying a transition requires a size study and adequate sampling; neither sketch is a verdict on this toy.

Two things make this hard. Small systems hide the hill: with only a few pieces, everything can flip at once, so even a sharp change looks like one broad hump. And the real test is whether the valley deepens as the system grows, which needs sizes nobody can simulate cheaply. Still, we set out a series of simulations that let us describe a sharp change of this kind in some detail.

Here is the author's idea, in one paragraph. First, who is asking, and why. She is a software engineer, not a physicist, and also a startup founder and a foster mom, so her recreational time is scarce and the computing bills are her own. The project is called RecreationalPhysics anyway. Her reason is simple: she wants to map her intuition about how reality is organized onto what humanity knows about the universe.

THE AUTHOR'S CLAIM Space is one settled arrangement of something deeper, which she calls X because nobody knows what it is. X was not chaos. It was a specific arrangement, stable for now, the way a reagent is stable until something gives it a push. The change that made space started at a seed, spread as a front, released a definite amount of energy all at once, which became matter and light, and left a remainder. Across the change the energy books balance.

That paragraph is a hypothesis, not a result, and nothing in a computer can prove it about reality. What a computer can do is take a model in which space is an arrangement of links, and ask whether a change of this kind can happen in it at all: whether there is a hill, how high it is, what comes down the far side, and what the far side is like. If the answer were no, the idea would have no home in that model. If the answer is yes, the idea gains a laboratory, and every number in it becomes a thing to check.

The point to keep: We cannot rerun the birth of space, so we build a toy in which space is an arrangement and ask what *kind* of change produces it. The author's idea needs the kind with a hill, a push, a release of energy and a remainder.

2 THE TOY

A published model of space made of links, and the one knob we add

PUBLISHED Our toy is built beside a published model. That model is called combinatorial quantum gravity, and it is the work of Carlo Trugenberger, Christy Kelly and Fabio Biancalana [1, 2, 3]. The published model is a network of dots and links with no positions at all: a dot has no place, only neighbors. Figure 4 shows every rule it has, and our toy keeps all of them. The mathematical word for such a thing is a **graph**; its dots are vertices, its links are edges.

Two names are kept apart from here on. **The published model** is theirs. **The toy** is ours: the published model with one of its prices turned into a knob, as described below. Wherever this paper says "the toy," it means ours, and the published one is always called the published model.

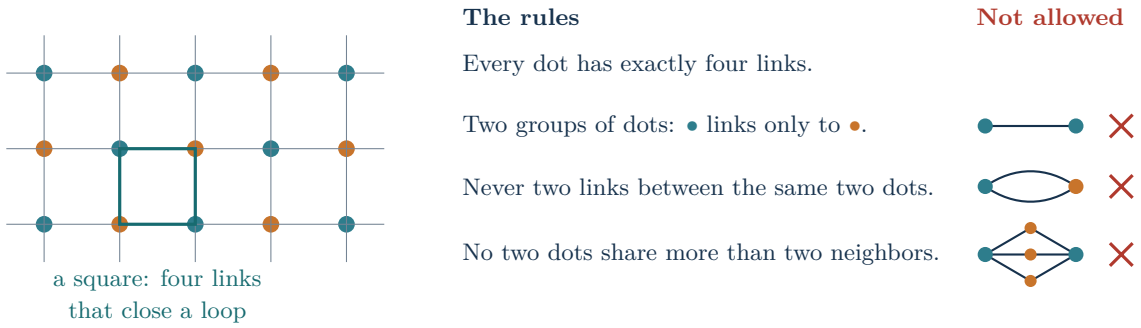


Figure 4. Every rule of the published model, all of which the toy keeps: dots, four links each, two groups, and the loops of four links called squares. There are no positions. Drawing the dots on a grid is a convenience; only the links are real.

The price list

PUBLISHED The published model gives every arrangement an energy, and the energy is worked out by counting squares. Compared with an arrangement that has one square for every dot, every missing square has a cost. A single link may carry at most three squares; a link carrying a third one has one square too many, and every such surplus square has a cost of its own. Those two prices are the entire rule. To be clear, this energy has no meters or joules in it. We will just call its amount a **unit**, and leave the conversion of units into physical energy for a later paper. Everything below is in these units.

PUBLISHED In the published model a missing square costs 16 units and a surplus square costs 4. With those two prices the energy is a measure of how curved the network is: a graph version of the curvature in Einstein's equations of gravity. That is where the published model's name comes from, a theory of gravity built by combinatorics, the mathematics of counting. The published work also studies a simpler version with the surplus price switched off, where a link can carry a third square for free.

The toy turns that one price into a knob. In the toy a surplus square costs 4λ units, where λ (lambda) is a number that can be turned up or down, so we call it **the knob**. With the knob at exactly $\lambda = 1$ the toy is the published model. With the knob at $\lambda = 0$ it is the published simpler version. Every other setting is our addition.

The whole point of the published program is to get general relativity, Einstein's theory of gravity, back

out of a network, so the curvature setting is the one that matters to it. At any other λ, however, the energy is still a perfectly good rule for a network, but it is no longer a curvature. So everything in this paper at $\lambda \neq 1$ is a close cousin of combinatorial quantum gravity, not the thing itself.

We have not found published work that sets the surplus price to any value above 1 [11]. In print, as far as we have found, are the two settings just described and, in the text of two of the papers [2, 12], a version in which a third square is forbidden outright. So, barring a gap in our search, the approach taken here and what it turned up are new.

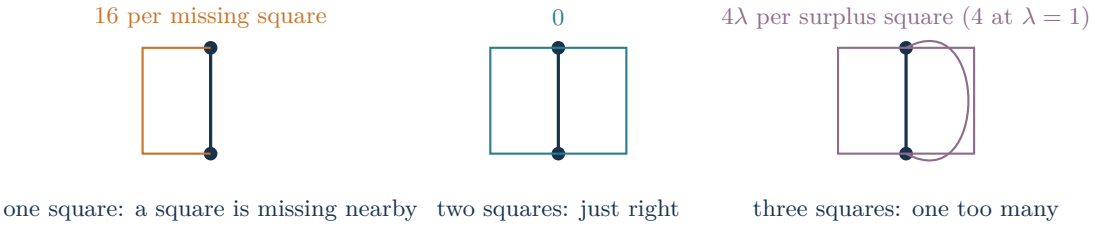


Figure 5. The price list, read link by link. EXACT The thick link carries one, two or three squares. Two is free. Fewer means a square is missing nearby, at 16 per missing square; a third is a surplus square, at 4λ, set by the knob. At $\lambda = 1$ the two prices make the energy a measure of the network's curvature; at $\lambda = 0$ a third square is free.

The one move

The network cannot leap to a new arrangement. PUBLISHED It changes by **swapping partners**: pick two links, Alice–Ben and Cara–Dan, and rewire them as Alice–Dan and Cara–Ben. Every dot keeps its four links and its group. Any swap that breaks the other rules is thrown out. That is the only move.

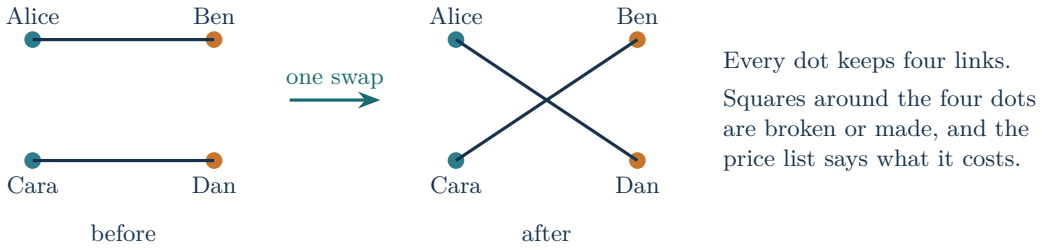


Figure 6. The one move the network has. PUBLISHED Two links swap partners. Everything that happens in the rest of this paper is a long chain of these.

Warmth, and the bath

PUBLISHED A run, of the published model or of the toy, is a long chain of such swaps, each one proposed at random and then accepted or refused. A second number, g, decides which. It plays the part of temperature, and we call it **the warmth**. A swap that lowers the energy is always accepted. A swap that raises it by ΔE units is accepted only some of the time, with a chance of $e^{-\Delta E/g}$, where e is a fixed number, about 2.7: the bigger the cost, the rarer the swap, and the warmer the network, the more often it goes through. Picture the network sitting in a bath that lends it energy for an uphill swap and takes back whatever a downhill swap gives off. The bath is endless: it never runs out and never fills up. Lowering g is cooling the network. Time in a run is counted in **sweeps**: one sweep is

two proposed swaps for every dot. Rules of this kind are the standard way to sample such a model; they are not a claim about how anything moved in real time.

The question the field asks, and reproduction as a starting point

PUBLISHED Start from a random tangle of links, cool it slowly, and watch. Does geometry condense out of the tangle sharply, with two humps, or smoothly? Published work reports continuous behavior for curvature-based models [2, 3] and first-order behavior in a related global-only model [4], where the network fragments. These are not identical experiments: the 2019 simulation forbids surplus squares, the later curvature action permits them, and the global-only study does not use the same fixed bipartite ensemble. With a third square free, cooling packs in as many squares as the rules allow, and the most tightly packed arrangements close up on themselves; Chapter 3 shows what those pieces are. Kelly's doctoral thesis [5] adds a hint for four links per dot: cooled and then warmed again, the network does not quite retrace its path, a small but persistent gap of the kind a sharp change leaves, though the thesis declines to call the evidence definitive.

MEASURED We checked published observables before extending the model.[1] Our simulations match the 2019 square-density curve [2] with a root-mean-square difference of 0.005 at the same size, match exact averages at sizes small enough to list every possible arrangement, and reproduce the sharp change into closed pieces with the surplus price off. Then we measured how much energy the change gives off as the tangle orders, at several sizes, with the rules written down first.

Setting	Dots	Energy given off per dot as the tangle orders
Surplus price off ($\lambda = 0$)	48	10.3
	64	11.3
	96	12.5
Full rule ($\lambda = 1$)	36	at most 1.65
	64	at most 1.51
	100	at most 1.30, and falling

MEASURED With the surplus price off, the energy given off is large and grows with size, but the cold state is a pile of closed pieces. With the full rule, there is one hump at every size we ran, and the estimated energy gap is small and shrinking. Settings of 1.25 and 1.5 gave the same picture. Those registered equilibrium tests were inconclusive; agreement with one published observable does not resolve the differences between the published ensembles.

The point to keep: The toy's energy is a price list: 16 per missing square, 4λ per surplus square. At $\lambda = 1$ the toy is the published model and its energy is a curvature, which is why that setting is studied. Out of randomness, at that setting, our tested sizes do not establish a first-order transition. A single histogram peak does not establish a continuous transition either.

[1] Our square-density curve differs from a 2025 figure [3] by up to 0.41 over part of its range. Given the different stated protocols, agreement with the 2019 curve does not validate the whole uncapped equilibrium curve. This reproduction issue remains open.

3 THE PHASE BEFORE

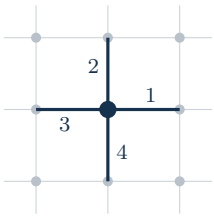
What if what came before space was not random?

The published question starts from a random tangle. This program asks a different one: what if what came before space was not random at all, but a specific arrangement, stable for now, with energy to give? That question is the focus of everything that follows. To ask it in the toy, we first need to say what space looks like in a network with no positions, and what the alternative to it could be.

Directions

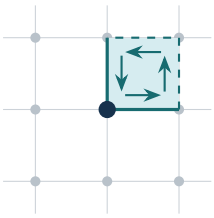
Space, at its simplest, is a set of large directions: directions you can keep walking in. Figure 7 shows how to find them in a network that has no positions.

1. Four links, six pairs



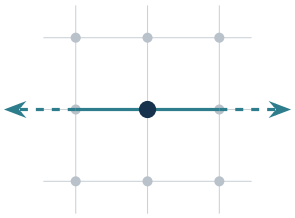
Every dot has four links. Four links make six pairs: 1–2, 2–3, 3–4, 4–1, 1–3 and 2–4.

2. Right angles: a square



A pair at right angles closes a square: out along one link, one more step, and back along the other. Four of the six pairs do this.

3. Straight through: a direction



A pair that runs straight through closes no square: follow it and you keep going. That is a **large direction**. Two of the six pairs do this.

Figure 7. What a direction is, in a network with no positions. EXACT Count the pairs at a dot that close no square, and call the count d. On a flat grid, $d = 2$.

A grid with no edges

A grid drawn on paper has edges, where the dots run out of links. The toy removes the edges.

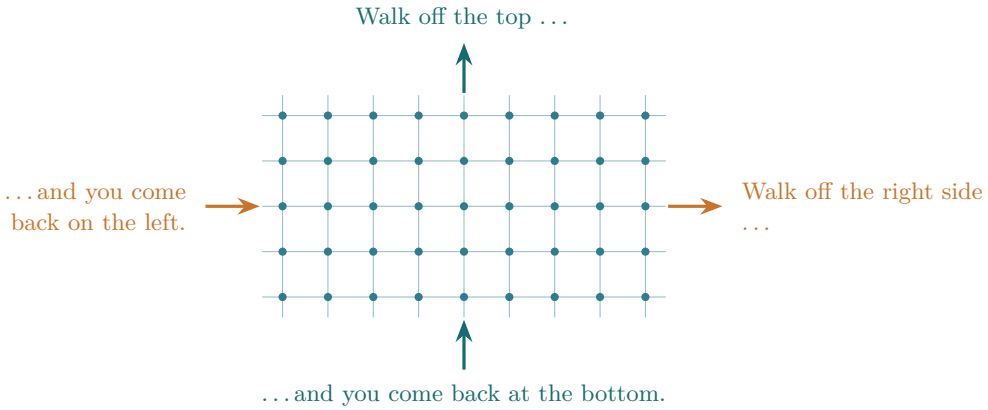


Figure 8. The toy's flat space. EXACT The grid wraps around both ways, like an old game screen whose edges join. It has no edges, so every dot has its four links, and when both ways around are long every dot reads $d = 2$. We call it **the flat sheet**. Mathematicians call any grid that wraps both ways a **torus**. These ideal wrapping grids are tori; arbitrary networks reached during a run need not be.

Curling

If a direction is not large, what else can it be? Small: curled up on itself.

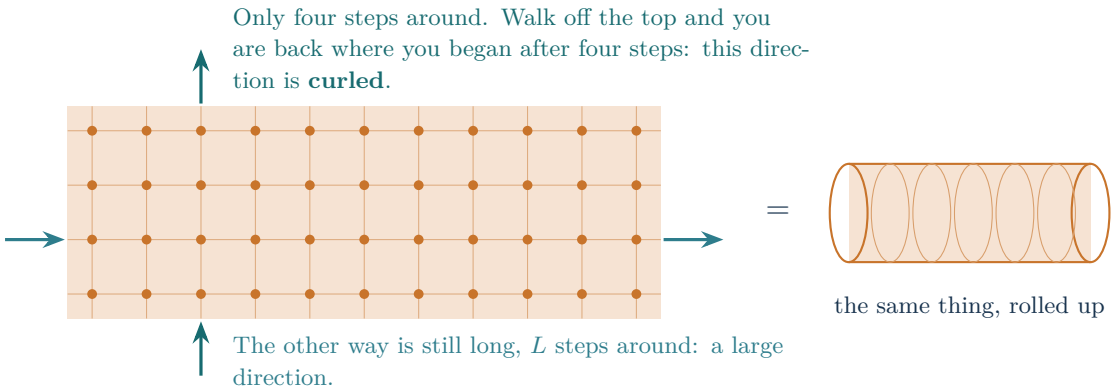


Figure 9. Curling one direction. EXACT Make one of the two ways around only four steps long. A grid four dots around and L dots along, wrapped both ways, is still a torus, the $4 \times L$ torus: the curled kind. We call it **the tube**.

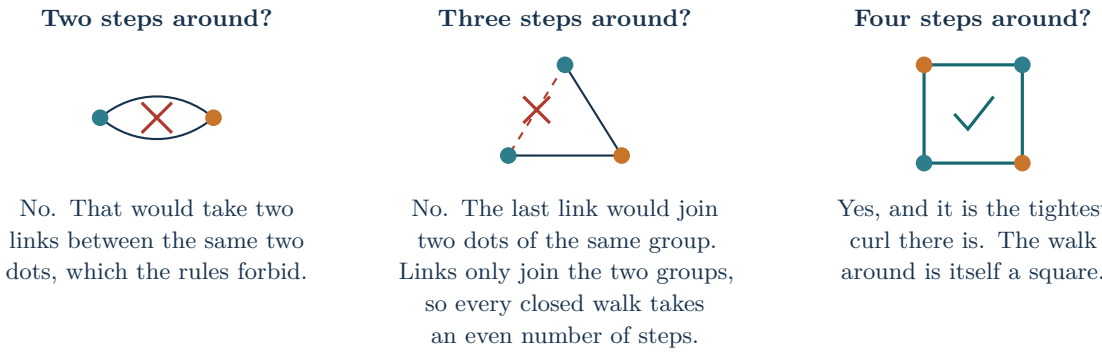


Figure 10. Why four steps around, and not three or two. EXACT

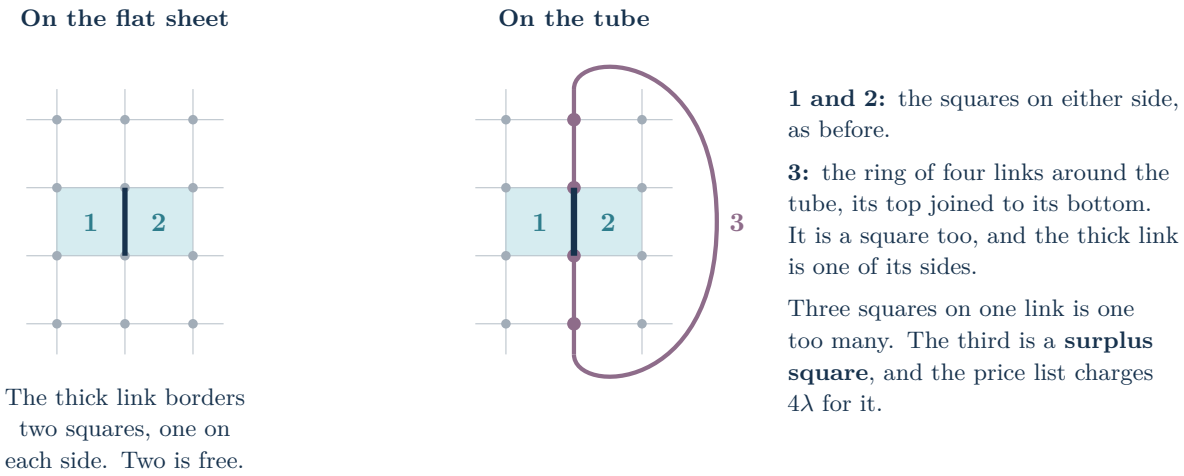
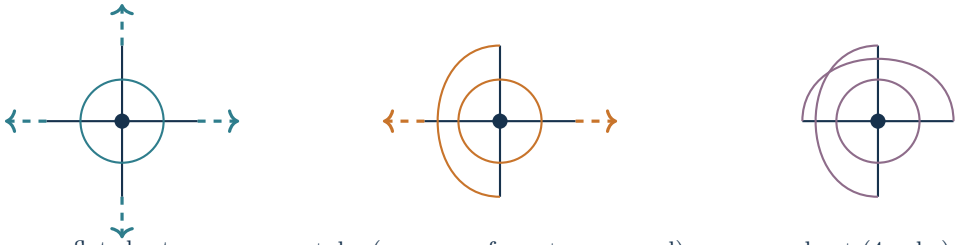


Figure 11. Curling puts a third square on a link. EXACT So curling a direction is exactly what the surplus price charges for. And because the way around is now a square, the pair of links that points that way closes one: the direction no longer counts as large, and every dot on the tube reads $d = 1$.

Curl both ways around to four steps and every dot reads $d = 0$: a closed sixteen-dot knot called the 4-cube. It is a torus too, curled both ways. Most of the closed pieces of Chapter 2 are these knots. Figure 12 puts the three arrangements side by side.

One dot in each arrangement. An arc marks a pair of links that closes a square. A dashed arrow marks a pair you can keep walking along. A loop marks a pair that is curled shut.

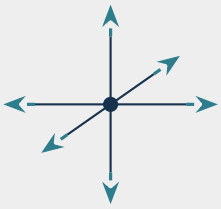


flat sheet
$d = 2$

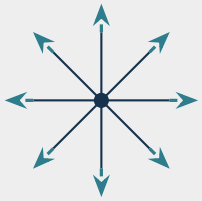
tube (one wrap four steps around)
$d = 1$

knot (4-cube)
$d = 0$

This paper: four links per dot, and the simplest change on the ladder: one curled direction opening (d from 1 to 2).



six links per dot: 15 pairs; 12 at right angles close squares, 3 run straight through
a flat grid has d=3



eight links per dot: 28 pairs; 24 at right angles close squares, 4 run straight through
a flat grid has d=4 (drawn flat: paper has only two)

Outside this paper.
Six and eight links per dot give three and four directions. A later paper takes them up. Nothing in this paper depends on them.

Figure 12. How many directions a dot has. EXACT Top: with four links per dot, the count d is the number of pairs of a dot's links that close no square, and it tells the flat sheet, the tube and the knot apart dot by dot. Grey box, outside this paper: the same count with six or eight links per dot allows three or four large directions, the subject of a later paper. The count is a signature, not a certified geometry: Chapter 6 says what it can and cannot vouch for.

The ladder

EXACT Put the three arrangements side by side: flat sheet, tube, knot. All three are tori, and we tell them apart by how they are curled: flat, curled, knot. Each curls one more direction than the one before, so we call them the rungs of a **ladder**. The price list of Chapter 2 gives each rung its bill, per dot:

	flat sheet	tube	knot
Directions curled	none	one	both
Squares per dot	1	$1\frac{1}{4}$	$1\frac{1}{2}$
saved on squares, at 16 each	0	4	8
Surplus squares per dot	0	1	2
paid for surplus, at 4λ each	0	4λ	8λ
Bill per dot: paid minus saved	**0**	$\mathbf{4(\lambda - 1)}$	$\mathbf{8(\lambda - 1)}$
at $\lambda = 1.25$, where a surplus square costs 5	0	1	2

Curling is a trade between the squares it gains and the surplus it pays for, and the knob decides which side wins.

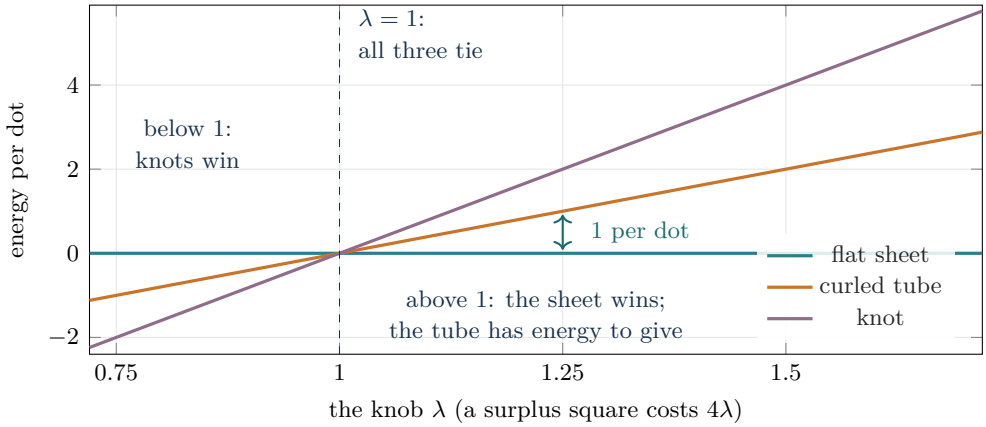


Figure 13. The ladder against the knob. EXACT Each line is one rung's energy per dot as the surplus price changes. At $\lambda = 1$, the published setting, the three rungs cost the same, so turning one into another releases nothing. Above 1 the sheet is lowest and the tube sits above it with a known amount to give: at $\lambda = 1.25$, exactly one unit per dot. Below 1 the ladder flips and a sheet would gain by curling into knots, which is why the network breaks into knots when the surplus price is off.

This is the first thing the toy teaches. EXACT At the curvature setting, curled and flat cost the same, so conversion between these ideal states releases no energy. Equal energies do not mean equal likelihoods or forbid a change. The knob must exceed 1 for this particular ideal energy gap to be positive. There, a tube ending at zero energy gives off exactly $4(\lambda - 1)$ units per dot. That is a reason to study this deformation, not a proof of a first-order transition or a physical reason why nature should choose it.

That release is what the rest of this paper is about, so it gets a name of its own. The author calls it **the burp**: the energy given off, and by extension the event that gives it off, a quiet wait, a start, a

spreading change and a definite release.

PUBLISHED The standard account of a stable-for-now state giving way calls its central calculation "the bounce" [8]. The two names point at opposite ends of the event. The bounce is about the start: it gives how often a seed of the new state appears, in a quantum system that tunnels *through* its hill. The burp is about the end: what comes out. And the toy has no tunneling. Its tube goes *over* the rim, with energy lent by the bath or handed to it as a push, never through it.

EXACT The changed rule is the published curvature energy plus $4(\lambda-1)$ times the surplus square count. The extra term vanishes when that count is zero. THE AUTHOR'S CLAIM The author proposes reading it as a *curling cost*: energy has to be paid to curl space tight, and is given back when it uncurls. She holds that as something to test. EXACT In the toy it is plain bookkeeping: a curled tube sits $4(\lambda-1)$ per dot above the flat arrangement, so uncurling gives that much back and curling again would have to pay it. OUR READING, UNVERIFIED What is not supplied is a derivation. At $\lambda = 1$ the rule is the published graph version of the curvature in Einstein's gravity; nothing known requires the extra term or fixes its size, so its meaning for nature is unverified.

Stable for now: is there a window?

Energy to give is not enough. Supercooled water has energy to give and sits there. For the tube to be a stand-in for X it must be stuck: leaving its equal-energy family must cost energy. EXACT At 18 dots the toy is small enough to list every allowed arrangement: there are 26, counting arrangements that differ only in the names of their dots as one. Among them, arrangements above the flat sheet from which every single swap costs energy exist for knob settings between 1 and 1.6, and for no larger setting. So the knob has a window, and a narrow one.

The stand-in

Our stand-in for X is therefore the tube. At $\lambda = 1.25$ it sits exactly one unit per dot above the flat sheet. It is a stand-in and not X: it has one large direction where space has three, its curled direction is always four dots around, and it lives in a network of a few hundred dots. What it shares with X is the shape of the idea under test: a specific, curled, stable-for-now arrangement with energy to give.

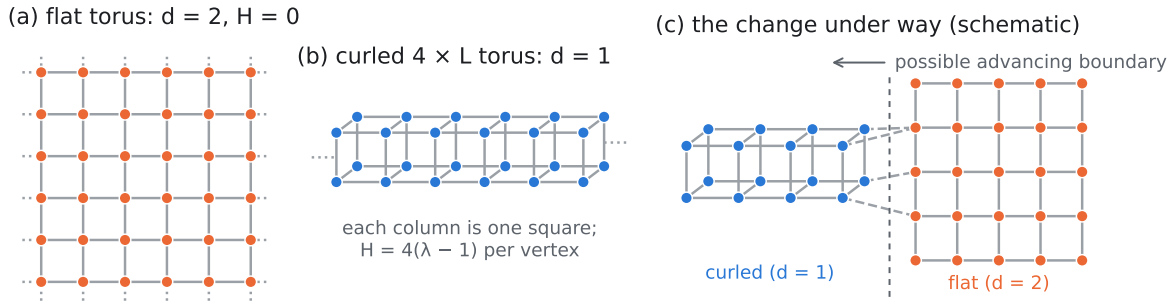


Figure 14. The two arrangements. (a) The flat sheet: a wrapping grid, two squares on every link. (b) The $4 \times L$ torus drawn as a tube: its short way around is a single square, so the links around each column carry three. (c) A sketch of the change, not a snapshot from a simulation. In the labels, H is the energy and a vertex is a dot.

The running example

Take a tube of 96 dots: four around, 24 along. At $\lambda = 1.25$ it holds 96 units above the flat sheet. That is the burp on offer. We will follow this tube through the rest of the paper so that the amounts stay concrete. It is a running example, not an invented history: the evidence comes from hundreds of repeated runs and, where possible, from counting every allowed move.

Now the question that decides whether any of this is a story at all. Ninety-six units are waiting to come out. What does it cost to start?

The point to keep: A direction, in the toy, is a pair of links at a dot that closes no square. Curling a direction four steps around turns the walk around it into a square, which puts a surplus square on its links. The tube has one large direction where the sheet has two, and sits $4(\lambda - 1)$ per dot above it: one unit per dot at $\lambda = 1.25$. At $\lambda = 1$ there is nothing to release, which is why the knob sits above 1, and why the toy is a cousin of the published model and not that model itself.

4 THE OBSTACLE

What does it cost to start?

Before any number, consider what the number could mean. If the push needed were tiny, a few units, then any warmth at all would set the tube off, and it would not be stable for now. If the push were as large as the burp itself, 96 units for our tube and more for a bigger one, then starting the change would cost what the change gives back, and the cost would grow with the size of the tube. Nothing small could ever start something large. A sharp change of the kind pictured in Chapter 1 needs a middle answer: a push that is small, and that does not grow with the size of the thing being pushed, so that a bigger tube releases a bigger burp from the same spark. A fixed push and a growing release is the condition for a runaway.

So: is the cheapest exit from a perfect tube small, fixed, or neither?

Pricing the cheapest exit

EXACT Start from a perfect tube and try every swap the rules allow, the one move of Chapter 2. Each one breaks some squares and removes some surplus, and the price list gives its cost. Most are expensive. Some cost nothing: they twist the tube around itself without changing any dot's neighborhood, so they do not count as leaving. Among the swaps that do leave, two kinds are cheapest. Call them route A and route B. Here are their bills at $\lambda = 1.25$, where a surplus square costs 5:

	Route A	Route B
Squares broken	2	4
charged, at 16 each	+32	+64
Surplus squares removed	4	10
refunded, at 5 each	−20	−50
The bill	**12**	**14**

Route A is the cheaper at this setting; the two tie at $\lambda = 4/3$, and B is the cheaper beyond. (For readers who want every exit listed with its formula, the technical paper [9] has them in Section II.)

The cheapest way out of the tube costs 12 units.

Twelve. Not 96, and not 2. For our 96-dot tube the push is an eighth of what comes out; for a 288-dot tube, which holds 288 units, the same 12 is a twenty-fourth. EXACT The cost of the cheapest single swap does not depend on how long the tube is, because a swap only touches a handful of neighboring dots; the burp does, because every dot contributes one unit. That is the middle answer the change needed, and it is arithmetic.

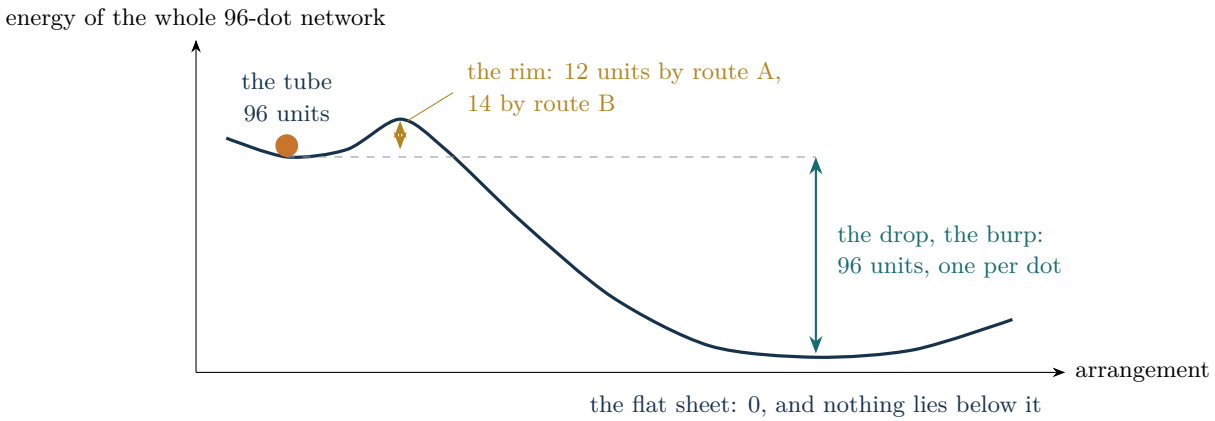


Figure 15. The landscape with its numbers filled in. EXACT The hollow is the tube; the rim is the cheapest swap out; the floor is the flat sheet. For our running example the rim is 12 and the drop is 96. Make the tube three times longer and the rim is still 12 while the drop is 288.

Is 12 really the floor?

One cheap exit from the perfect tube is not the same as no cheaper exit anywhere nearby. The free twists lead to other arrangements with the same energy, and one of those might have a cheaper door. EXACT So the calculation followed every free twist, at six sizes from 48 to 288 dots, until no new arrangement appeared. At each size there are exactly three reachable arrangements, counting arrangements that differ only in the names of their dots as one; every one has the same menu of exits, and the cheapest costs 12. This is a complete finite computation, conditional on the enumeration code being right, not a proof for every conceivable size: at a control size of 32 dots the pattern is different, which is a reminder not to extrapolate.

The test from the other side

Arithmetic says nothing below 12 can leave. Does 12 suffice? To find out, take the bath away and hand the tube an exact budget and nothing more (Figure 16), a standard way to simulate at a fixed total energy [7].

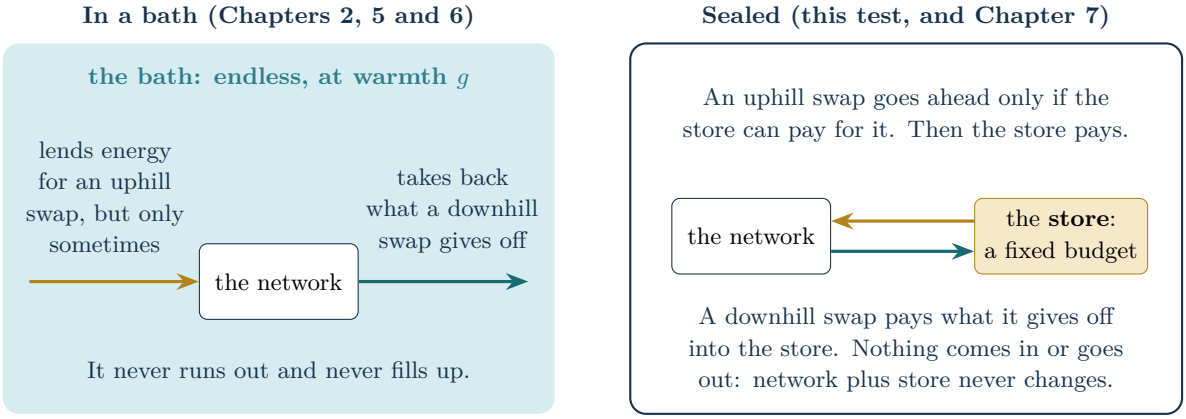


Figure 16. Two ways to run the toy. EXACT In a bath, energy for an uphill swap is always on offer, at a price in probability. **Sealed**, the network can spend only what its store holds. The store is not attached to any one place: it can pay for a swap anywhere in the network, so the push is a total budget, not a match held to one spot.

MEASURED, EXPLORATORY Try budgets of 4, 8, 10, 11, 11.5, 12, 13 and 16 units, eight runs at each budget at each of five sizes from 48 to 192 dots: 320 runs in all, each 12,000 sweeps long.

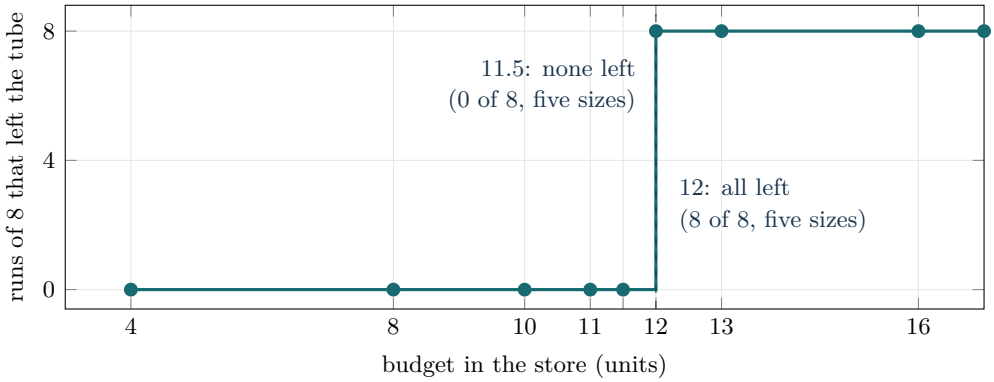


Figure 17. The staircase. MEASURED, EXPLORATORY The same curve at all five sizes, 48 to 192 dots: every budget below 12 gave zero escapes, every budget of 12 or more gave eight of eight. Pooled over the sizes, none of 200 runs left with a budget below 12, and all 120 left with 12 or more. The flag counted *leaving* the tube, not finishing the change.

EXACT The lower half of this staircase is a theorem of the toy at these sizes: with no exit cheaper than 12 and a store that cannot go negative, a budget below 12 can only pay for free twists. The upper half is an observation about this protocol and duration.[2] MEASURED, EXPLORATORY Together they say that the wall is local, exactly 12 units high at $\lambda = 1.25$, and the same at every size tried, while what waits behind it grows with the system.

Why we went here. This is the pivot of the whole paper. Suppose the push had grown with the size of the

[2]Why not run a hundred tubes at each budget instead of eight? Our reasoning, from the exact counts, is that it would add little. Below 12, leaving is impossible, which is arithmetic, and no number of runs can strengthen it. At 12 or more, the count of Chapter 5 says a perfect tube is offered, on average, three exits costing 12 in every sweep, and with 12 in the store the first one offered is paid for at once. A run that stayed put for 12,000 sweeps would itself be the surprise. The runs check that count; they are not estimating an unknown chance.

tube, so that a tube twice as long needed a push twice as large. Then no single spot could start the change: something would have to push the whole tube at once, and the picture of a seed and a spreading front would fail in this toy. We would have had to look for this kind of change in another one. A fixed, cheap push against a growing release is the condition under which something small can start something large.

The point to keep: Twelve units at $\lambda = 1.25$: the cost of the cheapest partner swap out of a perfect tube, counted exactly, the same whatever the tube's size, and confirmed from the other side by 320 sealed runs. The release grows with the tube; the push does not.

5 THE WAIT

How long before the door opens, and what kind of waiting is it?

A rim of 12 says what a push must be. It does not say when a tube left to itself will go. For that we put it back in the bath of Chapter 2 and let the network jiggle at warmth g. A swap that climbs the rim, costing 12, is accepted with chance $e^{-12/g}$: warmer, and the rim is cleared more often.

A lottery with counted tickets

Now the wait becomes a lottery, and every ticket can be counted. EXACT In one sweep the computer proposes two swaps for every dot, $2N$ in all for a tube of N dots, each one picked at random from every swap there is. How many of those proposals are exits?

Take route A. A tube of N dots has $3N$ different route-A swaps: three for every dot, so a longer tube has more ways out. But a longer tube also has more swaps of every other kind, and those grow faster, as N times N. So the chance that any one proposal lands on a route-A exit is 3 in $2N$, smaller for a longer tube. A sweep makes $2N$ proposals. Multiply the two and the length cancels:

$$2N \times \frac{3}{2N} = 3.$$

A sweep offers three route-A exits on average, whatever the size of the tube. A longer tube has more exits, each one is harder to hit, and with two proposals per dot in a sweep the two effects balance exactly. The same count for route B gives two.

Each offer of A is accepted with chance $e^{-12/g}$, each offer of B with chance $e^{-14/g}$. So the chance of leaving in any one sweep is $3\,e^{-12/g} + 2\,e^{-14/g}$, and the mean wait is one divided by that, with nothing fitted:

$$\text{mean wait} = \frac{1}{3\,e^{-12/g} + 2\,e^{-14/g}} \quad \text{sweeps.}$$

At $g = 1.5$, the warmth used for most runs below, that is 845 sweeps. Chemists know this shape as an Arrhenius law. Here both of its ingredients, the height of the rim and the number of tries per sweep, are counted from the arrangement rather than fitted to the data (the technical paper [9], Section II, gives the formula for any λ).

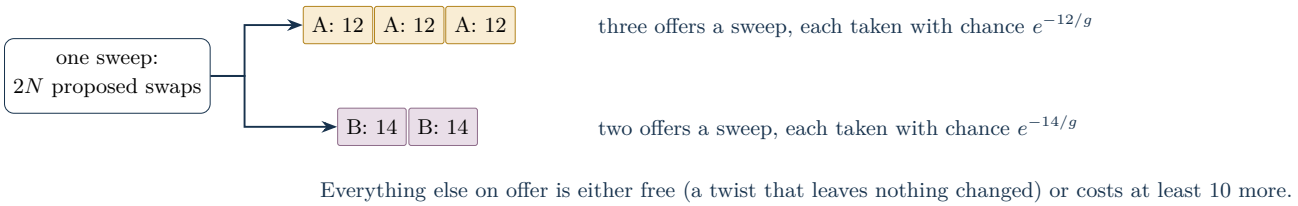


Figure 18. The lottery. EXACT The tickets are counted, not estimated: the perfect tube offers, on average, exactly three route-A exits and two route-B exits per sweep whatever its length. The chance of each being taken is set by the bath's warmth g.

Does the tube keep the appointment?

MEASURED, EXPLORATORY The first test ran twelve conditions at $\lambda = 1.25$: six warmths from $g = 1.4$ to 2.5, two sizes, sixteen tubes each, with measured waits from about 30 sweeps to about 2,500. The prediction written down for this test counted route A only, which makes the predicted wait a little longer than the full formula does: 994 sweeps at $g = 1.5$, where the full formula gives 845.

Averaged over the twelve conditions, the measured wait came within 2 percent of the predicted one. One condition at a time the match is looser: they range from 29 percent shorter than predicted to 62 percent longer (Figure 19). That is the scatter a lottery gives: with only sixteen tubes, their average wait lands, by chance alone, about 25 percent above or below the true average. The two sizes also differ. The 144-dot tubes sit near the full formula, and the 64-dot tubes waited longer. This exploratory test cannot say whether that is chance, which is one reason for the stricter test below.

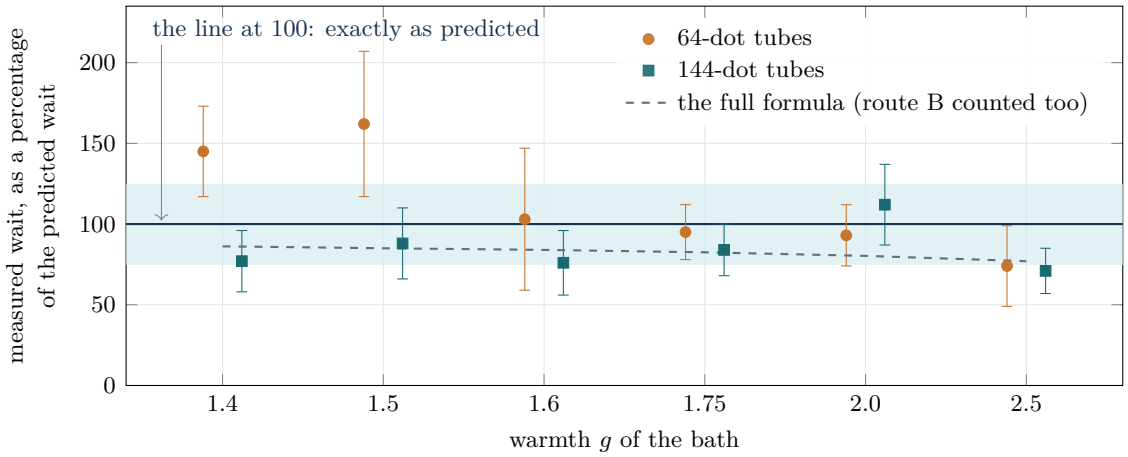


Figure 19. Twelve conditions against the count. MEASURED, EXPLORATORY Each mark is the average wait of sixteen tubes, shown as a percentage of the wait predicted from route A alone, so 100 means exactly as predicted. The bar through a mark is its uncertainty. The shaded band, 25 percent either way, is where such an average usually lands by chance when the prediction is right. The dashed line is where the full formula, with route B counted too, puts the prediction.

MEASURED A stricter, pre-registered test then counted every single accepted swap in 240 further runs across six settings of λ and size, and compared the first exits with the full formula, both routes. All 240 first exits were observed, and their mean arrived when the formula said it would in every cell, within the scatter expected from that many runs.

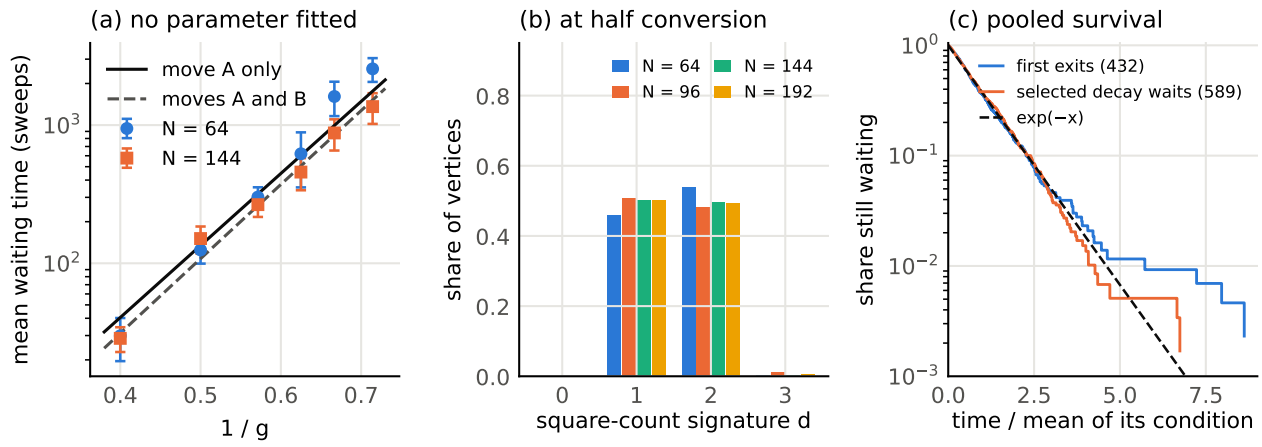


Figure 20. The wait, measured. (a) Mean waiting time against the inverse warmth $1/g$, at $\lambda = 1.25$; the lines are the formula with route A alone (solid) and A plus B (dashed), nothing fitted. (b) The share of dots at each direction count d halfway through the change, averaged over thirty runs per size: almost everything is at 1 or 2. (c) The fraction of tubes still waiting against time, in units of each condition's mean; a memoryless wait gives the dashed straight line.

What kind of waiting

If the chance of escape is the same on every attempt, the waits are **memoryless**. It is like rolling a die until a six comes up: twenty rolls without one do not make a six any likelier on the next roll, and a tube that has waited a long time is no nearer to leaving. One sign of a memoryless wait is that the waits spread out by about as much as their average. MEASURED That check was pre-registered, and it passed at every size in the main run. MEASURED, EXPLORATORY Checked afterwards, the whole distribution of first exits is consistent with that picture too, panel (c) above. The time to a *detected* change looked different, with a tail the first exits do not have. Chapter 9 finds out why.

MEASURED One more thing the move-by-move count revealed: the first swap out does not always lead anywhere. Think of that first swap as a door opening. About a third of the time the door closes again, and the tube goes back to being a perfect tube. Only 56 to 68 percent of first exits went on to a real change, with no trend in λ. The formula predicts when the door opens, not the departure.

A caution about the clock

OUR READING, UNVERIFIED The clock here counts attempted swaps, not seconds. One sweep offers any one neighborhood a way out about $1/N$ as often as a clock that gave every neighborhood its own fixed rate would. On that fairer clock the wait would fall as $1/N$, and a longer tube would go sooner. The toy has no time in it; which clock is "right" is not a question it can answer, and a later paper adopts the fairer one.

Why we went here. If the measured wait had disagreed with the counted lottery, something larger than a single swap would be holding the tube shut, and 12 would have been the wrong number. It agrees. The rim is the single swap, and the lottery is the whole explanation of the first exit.

The point to keep: Count the exits a sweep offers, multiply by the chance each is taken, and you predict the mean wait with nothing fitted. The tube keeps that appointment over a ninety-fold range of waits, and its waiting is memoryless, like rolling for a six. But about a third of the time the door closes again: an opened door is not a departure.

6 THE CHANGE

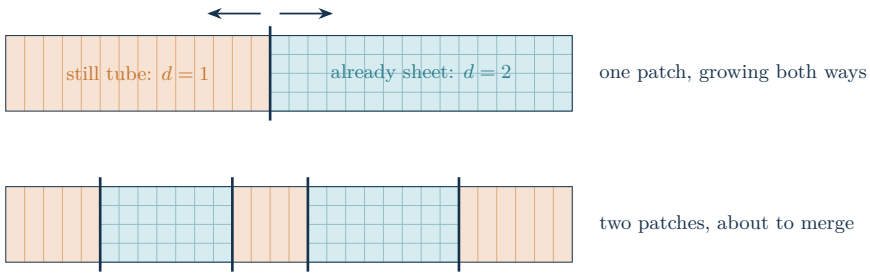
What takes shape on the other side

Chapter 5 was about the start: the first swap out, and whether it holds. This chapter is about what comes next: how the change spreads through the tube. It spreads fast.

The natural picture is a zipper. One spot of the tube opens, and the opening runs along the tube the way a zipper runs along a jacket. The slider, the tab you pull, is the boundary: behind it the tube has already opened into flat sheet, and ahead of it the tube is still closed. Physicists call a moving boundary like this a **front**. To find out whether the zipper picture is right, we need an instrument.

EXACT The fastest instrument is the direction count d of Chapter 3. Dots still in the tube read $d = 1$; dots on a flat sheet read $d = 2$; a melted, disordered dot reads 0 or 3 and up. The program marks every dot and also keeps a progress meter that runs from 0 (all tube) to 1 (all flat). It takes a snapshot of the whole network at three stages fixed in advance: when the meter first reads a quarter, a half and three-quarters. These are stages of the change, not times on the clock, and there is nothing special about them. They were fixed beforehand so that nobody could pick a moment because its picture looks persuasive. The predictions were scored at the half mark, where tube and sheet are most evenly matched and anything in between would have the most room to show.

MEASURED At $\lambda = 1.25$ and $g = 1.5$, thirty tubes at each of four sizes, run twice with fresh random numbers, and three predictions written down first: a memoryless wait; at least 80 percent of dots in one of the two signatures at the half mark; and at least 70 percent of the flat-like dots in one connected region, on average. All three held at every size in both runs. At the half mark 98.8 to 99.8 percent of dots read as tube or sheet. The other two snapshots were recorded but not scored. Read afterwards, they say the same: 98.8 to 99.7 percent at a quarter, and 98.5 to 99.5 percent at three-quarters. Something organized was happening, early, midway and late, with almost nothing in between.



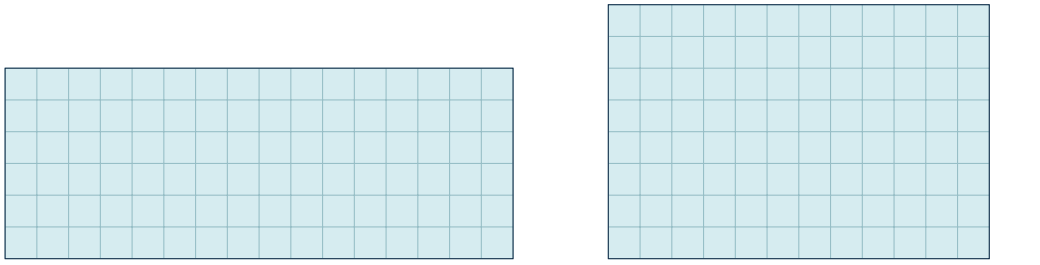
The tube is drawn unrolled: four dots tall, L dots long, its ends joined. After the patches merge, the two histories leave the same final picture.

Figure 21. What "one zipper" can and cannot mean. SKETCH, NOT DATA Both histories end with one large connected sheet-like region, and a snapshot cannot always tell them apart. The tube is a loop, so even a single open patch has two sliders, one at each end, moving apart.

What the count cannot vouch for

A jacket found open along one long stretch could have been unzipped from one spot, or from two spots whose openings ran into each other before anyone looked. So one large open region is weaker evidence than one zipper. MEASURED The averages passed the 70-percent rule at every size, but individual runs did not always. Of the thirty runs at 64, 96, 144 and 192 dots, 0, 2, 4 and 9 fell below it in the first set, and 0, 1, 3 and 5 in the second. A separate pre-registered test of a simple one-front law came out NOT ESTABLISHED: 21 of 59 tubes that started converted in several patches, mostly at the warmer bath. The honest description is that one connected flat-like region usually dominates by the half mark, and that the change often starts in more than one place: more than one zipper.

Nor does a count certify a geometry. EXACT A single swap from a perfect tube can produce dots that read $d = 2$ while their links carry one, two, two and three squares, which no dot on a flat sheet does. Two buildings can contain the same number of rooms and have different floor plans. And a grid six steps around and enormously long reads flat at every dot while keeping one direction short. The paper therefore calls $d = 1$ and $d = 2$ *curled-like* and *flat-like* signatures. When it says the curled direction opened, it means only that the four-step curl is gone, not that both directions grew large. The saved endpoints now let us test that distinction directly.



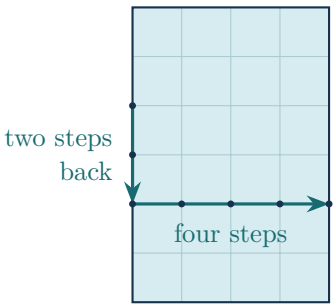
6 around, 16 along: 96 dots, every dot $d = 2$ 8 around, 12 along: 96 dots, every dot $d = 2$

Figure 22. Same count, different floor plan. EXACT Both grids wrap in both directions, cost zero, and read $d = 2$ at every dot. The left one still has a short way around. The signature alone cannot tell them apart; only measuring both ways around could.

A stronger look at the finished wiring

MEASURED, EXPLORATORY For the 120 saved final networks at $\lambda = 1.05$ in the original map, the wiring passes all four checks in Figure 25. A new, retrospective check goes further: it reconstructs every link as a square grid with its edges joined. All 120 are tori, but all remain narrow. The shortest loop around is six links at every size tested. A second route around, independent of repeating that first loop, is at least 16, 24, 36 and 48 links long at 64, 96, 144 and 192 dots respectively; in each case a loop of that length exists.

Think of the original four-around tube with its seam joined two places further along. That shift removes the four-step loop, but leaves a six-step loop. It is not a rectangular grid six dots across: the shifted seam still joins a grid with four times its length in dots. The energy has fallen and the local square pattern has changed, yet a second large direction has not appeared in these endpoints.



The right-hand endpoint and the upper left-hand dot are the *same dot* after joining the seam. Joining it adds no step.

A piece of the unrolled grid

Figure 23. Why the changed tube stays narrow. SKETCH, NOT DATA A seam shifted by two places joins the end of the four-step path to a dot two steps from its start. Four steps across and two back make a six-step loop. The long direction wraps too and is only partly drawn. The full reconstruction, not this sketch, establishes that six is the shortest noncontractible loop in the checked tori.

EXACT Shift the seam by two more places and the six-step loop goes as well, leaving eight; each further shift adds two. If the re-gluing kept happening, a second large direction would appear. But the toy's rule charges for loops of four and for nothing longer. Only the first shift releases energy. The later ones are neither rewarded nor charged, so nothing pushes them along.

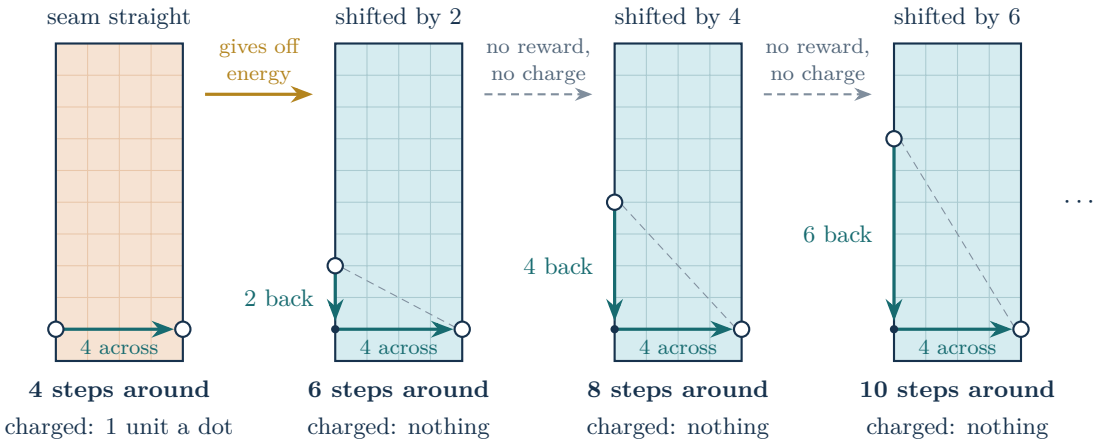


Figure 24. One shift pays; the rest are free. EXACT Each strip is a piece of the unrolled tube, four dots across. The two rings in a strip are one dot: the seam joins the end of a four-step row to it. With the seam joined straight, the row closes on itself in four steps. Each shift of the seam by two places sends the row's end two dots further along, so the shortest way around grows by two: six, eight, ten. Only loops of four are charged (one unit a dot at the setting most runs use), so the first shift gives off energy, and the later ones neither cost nor repay anything. The widths were checked up to twelve around, on tubes of 96 dots or more.

A word about "flat." Here it means flat at every dot: every link on exactly two squares, at zero energy. A tube six around is flat in that sense and still narrow, so it is not what most people would call flat. Does being flat at every dot at least make a wide shape likely? EXACT Counted as ways of wrapping the grid, yes: of the wrappings of 192 dots that are flat at every dot, about one in twelve is six around, and the typical one is twelve around. MEASURED, EXPLORATORY But the runs do not choose among them at random. Every move out of such an arrangement costs energy, at least 20 units from a tube six around and 32 from a wider one at the setting most runs use, so in a cool bath the network stays at the first one it reaches, and from the tube that is the nearest: six around.

MEASURED, EXPLORATORY The newer map in Chapter 9 makes another distinction necessary. Among its 3,521 saved zero-energy endpoints from runs that reached the three-quarter conversion mark, 3,446 are narrow tori with a shortest loop of six links; 73 are disconnected surfaces; two are Klein bottles, surfaces whose orientation reverses on going around them. The two runs outside that selection also eventually reached zero energy, but are not included in these counts. "Zero energy," "locally flat" and "one torus" are different statements. The supplementary geometry check records the selection and reconstructs the torus wiring rather than inferring it from width alone.

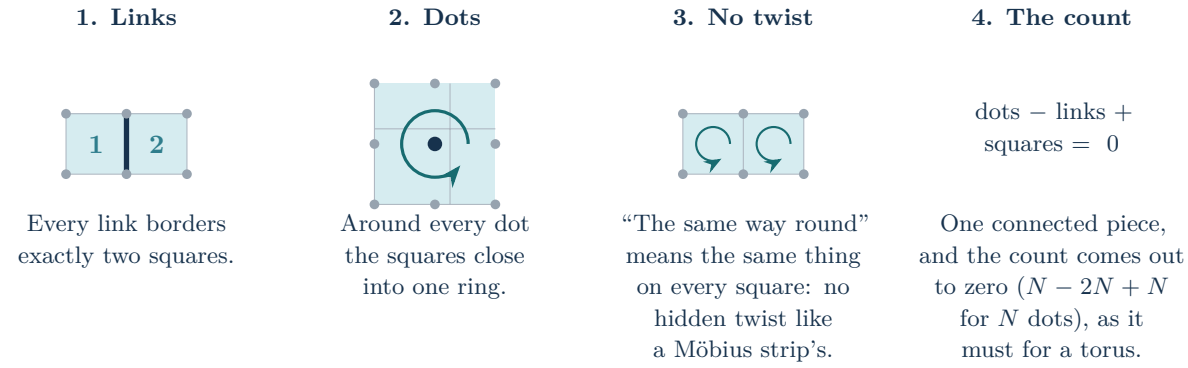


Figure 25. Certifying a torus from its wiring. EXACT Four checks, run on each of the 120 saved networks after filling in every square. A network that passes all four is a torus, whatever its dots' direction counts say.

The energy comes out exactly

MEASURED When a tube finishes at zero energy, the energy given off is exactly one unit per dot at $\lambda = 1.25$: the arithmetic of Chapter 3, now measured. In 24 to 28 of 30 runs per size the release was within one percent of that. The rest settled on sheets with small defects, 8 to 45 units' worth read from their wiring, and held back that much. The ideal gap is exact; how much of it comes out depends on where the network stops.

Why we went here. A snapshot during conversion tests whether the local signatures coexist and whether one connected patch dominates. Those are useful, limited questions. They do not by themselves establish a single advancing front, global opening or a thermodynamic first-order transition.

The point to keep: A quarter, half and three-quarters of the way through, about 99 dots in 100 read as tube or sheet, and one connected sheet-like region usually dominates. That is sharp in the sense written down beforehand. It is not proof of a single zipper. The original 120 finished tori remain narrow: their shortest loop is six links at every tested size. Local flatness has not produced two large directions here.

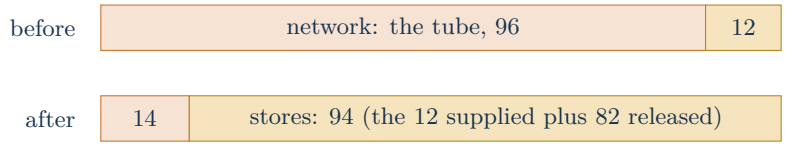
7 CLOSE THE LID

Where does the energy go when it has nowhere to go?

Apart from the budget test of Chapter 4, every run so far sat in the bath, and the bath carried the released energy away as fast as it appeared. If a change like this made all of space, there was no outside for its energy to go to. So seal the system again, as in Chapter 4, and watch what the burp does to its own surroundings.

EXACT The sealed system of Chapter 4 had one store. Here there are C stores beside the network, each holding a non-negative number of units. Every proposed swap picks one store at random: an uphill swap is refused unless that store can pay; a downhill swap deposits its release there. Nothing enters or leaves, so network plus stores is constant to the unit, and it was checked to be constant in every run.

Return to our 96-dot tube, holding 96 units. Put 12 more in a store and seal the lid. The account reads 108. Suppose the change runs and leaves one 14-unit scrap in the sheet, as Chapter 8 will show it often does. The network has gone from 96 to 14; the stores have gone from 12 to 94.



Both bars are 108 units long. Nothing is lost; it only moves between the network and the stores.

Figure 26. The ledger for the running example. EXACT The bookkeeping is exact in every sealed run; the particular ending shown here, one 14-unit scrap, is an illustration of a common outcome, not a promise about any one run.

The surroundings can spoil the ending

Here is the twist. The released energy is now in the stores, and the stores pay for uphill swaps. Energy that has nowhere to go can buy swaps that break the very squares the change has just made. How much damage depends on how many stores there are. PUBLISHED Many stores share the release thinly, like a large cold room; few stores concentrate it, like a small hot one.

MEASURED Pre-registered: tubes of 64, 96 and 192 dots, seven choices of C from a single store to two per dot, twenty runs at each setting, 420 runs. The outcomes were classified by rules fixed beforehand: sheet-like if at least 90 percent of dots read $d = 2$ with almost no melted dots; melted if 15 percent or more read as disorder; stalled if both orders sat side by side with nothing melting; in between otherwise.

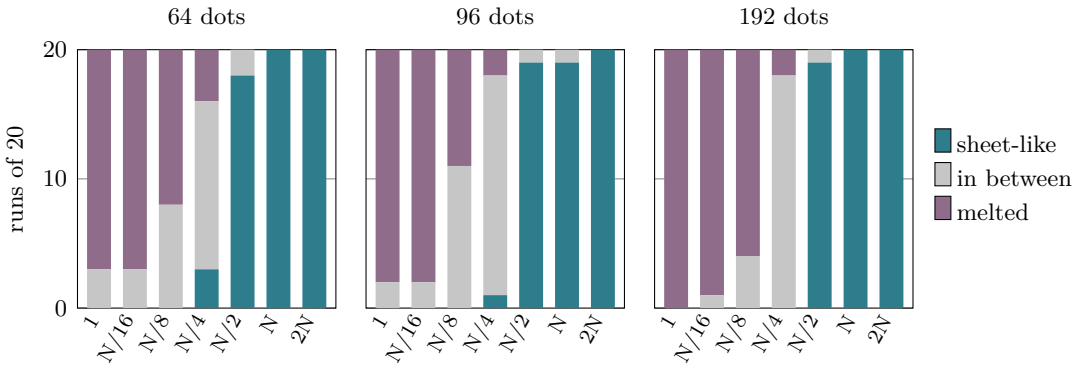


Figure 27. Bonfire or melt. MEASURED The horizontal axis is the number of stores C, as a share of the number of dots N. With few stores the release damages the new sheet (the registered label is "melted"; the next section says what that can mean); with many it leaves a clean sheet. The switch-over lies between one store per four dots and one per two at all three sizes, and no run ended stalled with both orders side by side. Totals: 179 sheet-like, 87 in between, 154 melted.

MEASURED A bonfire with a threshold, as the pre-registered verdict put it. Where the heat has room, the change feeds itself to completion; where it does not, the heat damages what the change made. Melted, in this classification, means that at least 15 percent of the dots have lost their neat two directions. The switch-over moves in step with size, as a release of one unit per dot shared among the stores predicts. OUR READING, UNVERIFIED A rough energy budget puts it near one store per three dots; the measured crossover sits between one per four and one per two, which is as fine as the grid can say. That estimate assumes the stores settle into a thermal balance, and individual store balances were not saved, so it is a budget, not a calibrated thermometer.

MEASURED No run ended stalled with tube and sheet side by side and nothing melting in 420 runs of 30,000 sweeps on this grid of C. That finite observation does not establish that no coexistence temperature exists. Such a claim would require free-energy and equilibration evidence that these sealed runs do not supply.

What "melted" can and cannot mean here

EXACT A sealed box holds only the energy it started with: at $\lambda = 1.25$, one unit per dot, plus the push. A missing square costs 16 units. So the burp's own heat can break at most about one square in sixteen, however few stores there are. MEASURED, EXPLORATORY Read again from the saved runs, every one of the 154 networks labelled melted still had 94 to 100 percent of its squares. That is damage: warm fabric with flaws in it. It is not the random tangle the published model starts from, which has about an eighth as many squares as flat fabric. The burp cannot melt its own product. An earlier draft of this chapter said that it could.

Does trapped heat make the product wider?

MEASURED Pre-registered, with the author's prediction that it would: tubes of 96, 192 and 384 dots at $\lambda = 1.25$ and 1.40, sealed with two stores per dot down to one per four, then drained of their heat, cooled, and read from the wiring. It did not. With one store per dot or more at $\lambda = 1.25$, every one of 144 tubes ended narrow, six around, and with half a store per dot most still did. With a third or a quarter of a store per dot, or at $\lambda = 1.40$ with any amount of room, most ended as a *patchwork*:

good fabric almost everywhere, six around in one stretch and eight or more in the next, with flaws frozen in where the stretches meet, and often in two or more separate pieces. None of 720 end states was one clean torus wider than six. The share that was mostly sheet-like fell as the tube got longer, from one in five at 96 dots to one in sixty at 384 ($\lambda = 1.40$). For the largest size the shares were written down beforehand from the two smaller ones: they held at nine of ten settings and missed at one. OUR READING, UNVERIFIED Trapped heat is flaws in many places at once, each re-stitching its own neighborhood. A clean re-gluing needs one edge moving alone.

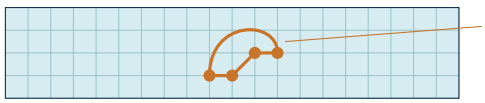
Why we went here. What the burp turns into, and whether the energy books balance across the change, only have meaning in a sealed system. In a bath the burp leaves and nothing can be asked of it. The sealed box is where the balance becomes a measurement.

The point to keep: Sealed, the release has to go somewhere, and where it goes decides the ending: into enough room and the sheet completes, into too little and the heat damages the new space. The books balance to the unit in every run. Nothing stalled half and half.

8 WHAT REMAINS

A scrap of the old arrangement

Almost every sheet that formed in a cold sealed box kept something: four dots of the old curled arrangement that the change swept past without opening. A remainder raises one question first: does it grow with the space? A remainder that grows in step with the space is a fixed share of everything, big enough to be some of what a universe is made of. A remainder of fixed size is a speck. The prediction, written down before the measurement, was that it grows.



the scrap: a closed loop of four dots, $d = 1$, lying along the old tube; 14 units at $\lambda = 1.25$

the finished sheet, drawn unrolled; its ends are joined

Figure 28. The usual leftover. EXACT Read from the wiring: a closed loop of four dots with one extra square and six surplus squares, costing $24\lambda - 16$ (14 at $\lambda = 1.25$), its points spanning one to four of the tube's original columns and lying along the tube. Earlier notes called it a ring around the tube; the wiring says otherwise. Occasionally a different, 20-unit twist appears instead.

MEASURED Eighty fresh runs in the coldest box, two stores per dot, pre-registered: tubes of 64, 96, 192 and 288 dots. The count of scraps per finished sheet was 0.90, 0.85, 1.05 and 0.95, with a spread of about 0.2 to 0.4 either way across the twenty runs at each size. A straight line drawn through them is flat, within that scatter. About one scrap per box, however long the tube.

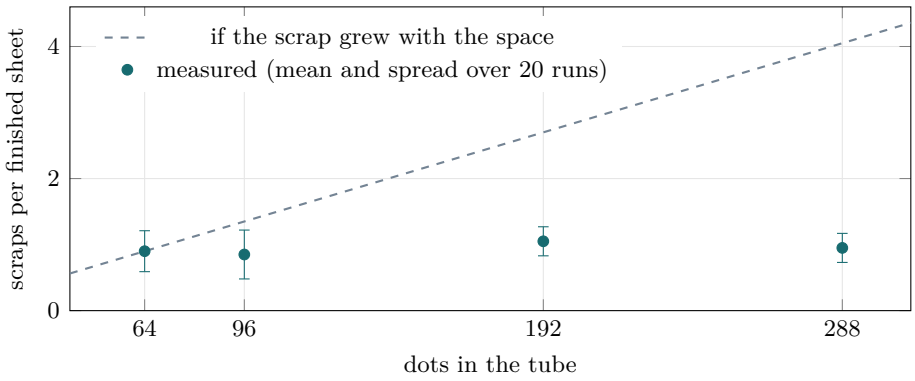


Figure 29. One scrap, however large. MEASURED The dashed line is what a leftover growing in step with the space would look like; the measured counts sit flat. The prediction that it grows, written down beforehand, failed.

Where does it sit, and why one?

It is tempting to call the scrap the place where two fronts met and left a seam. MEASURED That was pre-registered and tested: the scrap sits neither opposite the start nor at it, and a randomization check that moves it to every position along the tube finds nothing special about where it landed. We do not know why the count is about one in this protocol. The technical paper [9] says so under "not claimed."

What the scrap does afterwards

This paper stops at the count. Three questions follow from it, and they are the subject of the author's next paper: does a scrap last, or heal away, when the sheet around it is kept warm or is cooled; does a change that starts in several places at once leave more of them; and if so, is the leftover a speck or a share. The short version: how many scraps survive depends on how the change started and how fast the sheet cooled, and more than one prediction written down beforehand failed.

Why we went here. The change leaves a remainder in almost every run, at every size. The pre-registered test asked the one question that decides what kind of thing it is, and the answer, fixed size, is not the one the author predicted. It is reported as prominently as the results that went her way.

The point to keep: The change leaves about one four-dot scrap of the old arrangement per sheet, 14 units, whatever the size of the tube, sitting anywhere along it. From a single seed the leftover is a speck. What it does afterwards, and what several seeds leave, is the next paper's question.

9 THE MAP

Turning the knob

Everything so far was at one setting of the knob, $\lambda = 1.25$, chosen in advance. A single setting proves little: a result that only holds at one carefully chosen number is the kind of thing the project's rules forbid. So the whole stretch of the knob was run, from 1.05 to 1.45 in steps of 0.05, at four sizes, thirty tubes per cell, with the rules fixed first. Then it was run twice more with 120 tubes per cell and fresh random numbers. Each run carries a detector that decides when the change has happened. It watches the tube for its first 200 sweeps to learn what an unchanged tube looks like, and only then starts checking, so it cannot record a change earlier than that. The end of this chapter returns to what this costs.

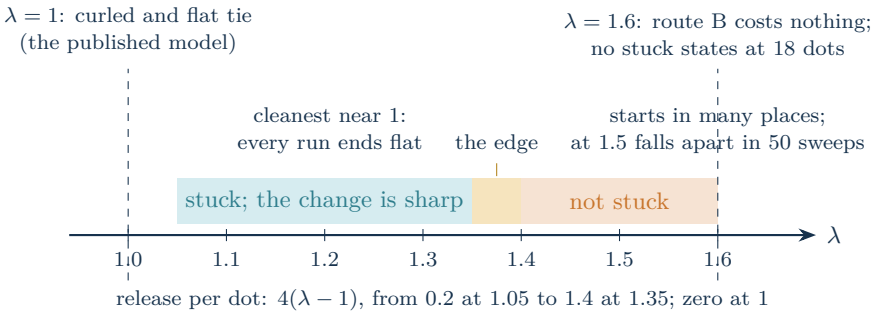


Figure 30. The stretch of the knob that was run. MEASURED Shaded, where the tube is stuck (more than half the tubes still at least three-quarters tube after the 200-sweep watch) and the change keeps its two-signature, dominant-region character. The edge between 1.35 and 1.40 is where the wait drops below the watch, not where the hill disappears: route B still costs energy until 1.6. EXACT

MEASURED Three things the map shows. First, the tube is stuck at every setting from 1.05 to 1.35 and at none from 1.40 up, and wherever it is stuck the change keeps the character of Chapter 6: 95 to 100 percent of dots in one signature or the other at the half mark, one connected region holding most of the flat-like dots. Second, the mean wait follows the counted lottery of Chapter 5 over a hundred-fold range of waiting times with nothing fitted, drifting above it only where the 200-sweep watch cannot record shorter waits, and running 15 to 43 percent long near $\lambda = 1$, where the direct move-by-move count finds no excess. Third, the change is cleanest near the published setting and gets messier as the knob turns: at 1.05 every one of 120 tubes ended as a perfect flat torus, at 1.25 about 88 percent did, at 1.35 fewer than half, and beyond the edge the change starts in several places at once in most runs and ends mostly on a sheet with defects.

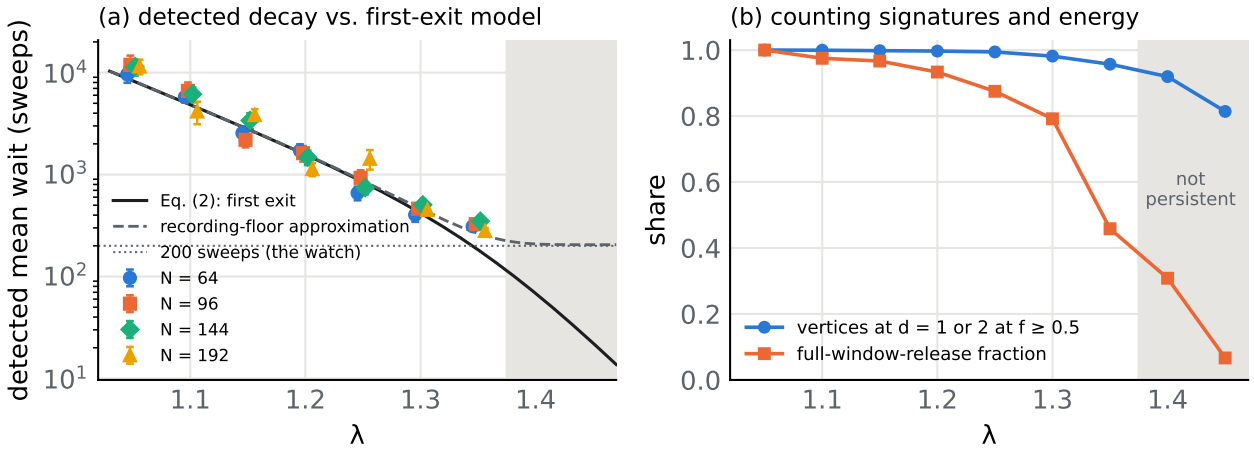


Figure 31. The map, measured. (a) Mean detected wait against λ at $g = 1.5$, four sizes; the solid line is the counted lottery for the first exit, the dashed line the same prediction as the 200-sweep watch would record it; neither is fitted. Shading marks where the tube is not stuck. (b) Pooled over sizes: the share of dots in one signature or the other at the half mark, and the share of runs that released the full amount.

OUR READING, UNVERIFIED There is a trade here. A larger λ means a larger burp per dot but a less patient X and a messier birth; a smaller λ means a cleaner change that releases less. The ideal release is $4(\lambda - 1)$ per dot, attained only at a zero-energy endpoint, and vanishes at the published setting, so this family of models has no setting with a large release and a quiet, clean X at once. PUBLISHED At the published setting itself, the published model's author has proposed that patches of the network can sit stuck in a different arrangement [6]; the questions asked here could be put to those patches too.

Three verdicts, all inconclusive by the letter

MEASURED Each of the three maps was scored against rules fixed before it ran, and each failed one of them: the first on a band for the spread of waits that, with thirty tubes, turned out to be not much wider than ordinary run-to-run scatter, so a cell could fall outside it by chance alone; the second on an energy check that compared a window average with an exact final energy and was tripped by the sheet's own thermal jitter; the third on the spread of waits again, in four cells, two of them on a single tube recorded as waiting 24 and 76 times the predicted mean (neither of those two was a wait, as the end of this chapter shows). Three runs, three sets of rules not satisfied. Each verdict stays inconclusive. Later diagnoses explain specific failures; they do not retroactively turn those tests into passes.

The long waits that were not waits

MEASURED One loose end from these maps looked alarming for a while. The third map recorded two tubes waiting 24 and 76 times the predicted mean, which a memoryless wait all but forbids. A dedicated pre-registered test of 10,000 tubes then counted too many long waits and returned the verdict TWO POPULATIONS. It also found a fast share, 12 to 23 percent of tubes that seemed to change at once, and 190 tubes the detector never caught.

Much of this apparent anomaly came from the detector (Figure 32). For its first 200 sweeps the detector only watches, to learn how much a resting tube jiggles, and then it sets its bar at three times that jiggle. A tube that changes *during* the watch teaches it that large swings are normal.

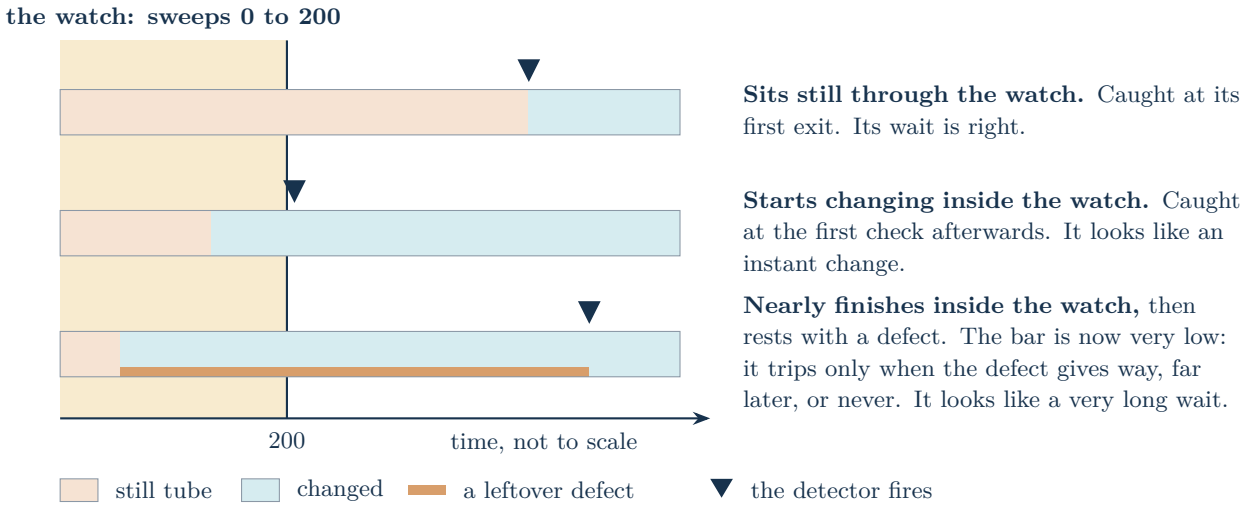


Figure 32. Three tubes, one detector. EXACT The detector's rule is the same for all three; what differs is what each tube did while the detector was still watching.

MEASURED, EXPLORATORY Every run saved two readings that do not depend on the detector's bar: how far the tube had converted when the watch ended, and when a quarter of it had converted. Read afterwards from those saved rows (a reading made after the fact, not a pre-registered test), the puzzle comes apart. The 190 tubes never caught had all finished at least three-quarters of the change inside the watch. The two extreme waits were tubes already 75 and 88 percent converted when the watch ended, resting on a sheet with defects. The fast share is tubes whose first exit came inside the watch, about as many as the counted lottery says there should be. And the test's yardstick for a typical wait, taken with the fast share included, came out too short, which pushed ordinary long waits past its cutoff.

The verdict stands as scored; these checks explain most of the apparent anomaly. Three long waits in one group of 4,000, at $\lambda = 1.30$ and 64 dots, were not explained this way, so we replayed that whole group move for move from its saved starting numbers, this time recording everything, with the rules for reading it written down first. MEASURED One of the three was the detector again. The other two looked like perfect tubes every time the replay looked, which was once every five sweeps. So we replayed them once more and looked after every single sweep. MEASURED, EXPLORATORY One had sat, perfect, for about twelve times the average wait. The other had slipped out for three sweeps between two looks and come back, about seven average waits in, so its long wait was the spacing of the looks. At no sweep did any tube hold more squares than a perfect tube, which a second curl would need. MEASURED Neither written guess about the three held (a hidden second curl; the detector three times over). Timed with no detector at all, the 4,000 first exits follow the counted lottery to within 7 percent, and one comes later than ten average waits, which chance allows about one time in six.

MEASURED One more run, with its rules written first, closed that 7 percent. On 4,000 fresh tubes we counted the first exit move by move and, in the same tubes, by a look every five sweeps. Counted by the move it lands on the count: 1.005 times the predicted wait, give or take 0.016. The looks read it about 10 percent late, because about one first exit in thirteen comes back before the next look. So the 7 percent was the looking, not the lottery.

MEASURED The same run settled an older puzzle at the lowest setting of the knob, $\lambda = 1.05$, where three tubes among 200 had waited about nine times the average. Replayed from their saved starting

numbers, each was a tube that simply sat: it was offered its ways out at the usual rate the whole time and took none. Then 3,000 fresh tubes there, counted move by move, showed no detected excess of long waits: one wait past eight averages, where chance expects one. The observations are consistent with ordinary random variation; a finite sample cannot rule out every rare alternative.

Why we went here. The project's first success condition is no cherry-picking: a family of rules may be explored, but only along knobs fixed in advance, and every setting run is published. The map is that rule applied, and it also locates where the stand-in for X stops being stable for now.

The fourth map and its held-out size

MEASURED A fourth registered map records the first exit move by move and uses 120 replicas per setting. All 24 combinations of $\lambda = 1.05$ through 1.30 and 64, 96, 144 and 192 dots pass its specified waiting-spread, signature and connected-patch tests. Predictions were frozen before reading the larger, 288-dot runs. All six window settings meet those prediction intervals and pass the conversion criteria there. At 1.35, the connected-patch criterion fails at 192 and 288 dots.

The pass has a precise scope. At 288 dots and $\lambda = 1.30$, the mean share in the largest patch is 0.751 against a cutoff of 0.7; only 65 of 120 individual runs exceed that cutoff. Passing a test of the spread of waits is not proof that the whole waiting-time distribution is memoryless. MEASURED, EXPLORATORY Applying the earlier map's residual-wait criterion to these same new rows also passes all 24 smaller-size window cells. The fourth map changes the sample and the diagnostic as well as the clock; its success cannot be attributed to the clock repair alone.

The point to keep: The fourth map passes its registered criteria over the tested window, including the held-out size. The earlier three remain inconclusive. This supports a bounded result about escape and conversion under the specified rules. It does not establish a thermodynamic first-order transition, and the reconstructed zero-energy tori still have a short way around.

10 THE EDGES

Where the simulations stop, and what lies past each edge

Every knob the toy has was turned only so far. This chapter is the honest map of the box: for each knob, what was run, why it stopped there, and what is on the other side, with the kind of knowledge marked. "Unknown" means no one has looked.

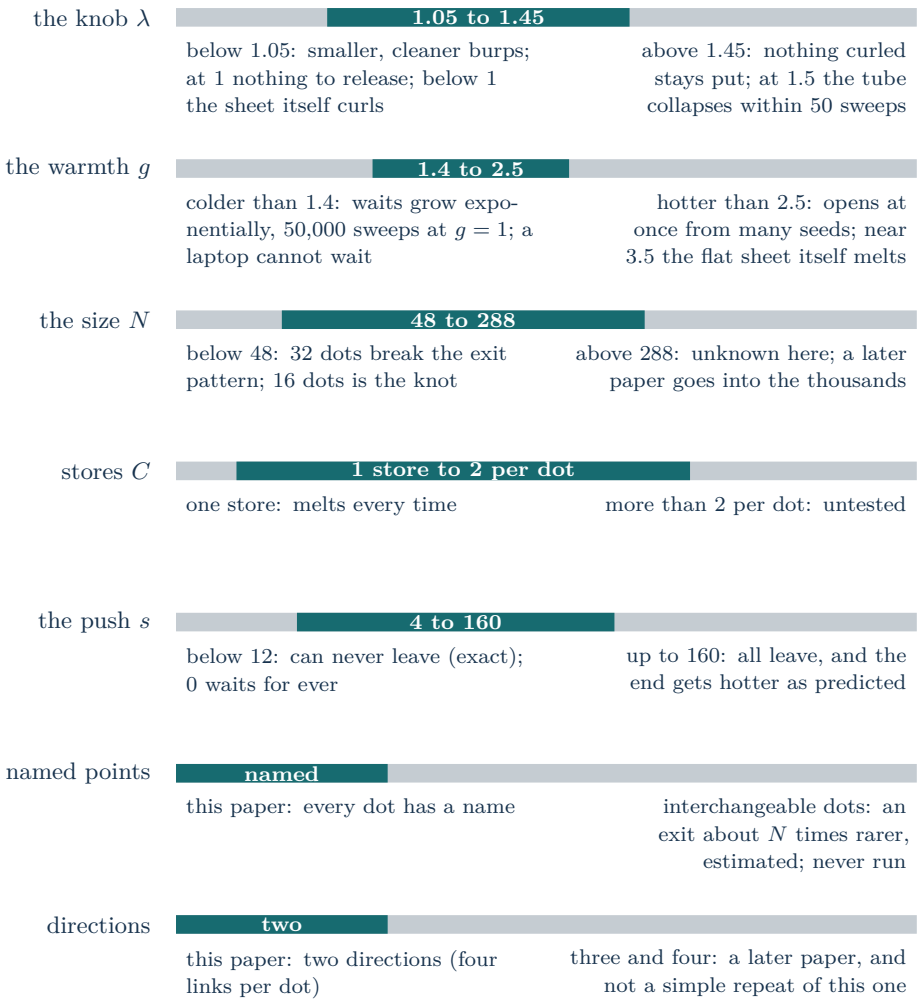


Figure 33. The box, knob by knob. Each teal bar is the stretch of a knob where tubes were actually simulated for this paper; the gray is the knob's range as far as anything is known about it; the notes say what lies past each edge. You can turn the first five knobs yourself in the interactive companion to this figure, The Tube's Curling Ladder: its exact numbers follow you to any setting, and its measured results appear only inside the teal bars. Details and the kind of knowledge behind each note follow.

The knob λ: 1.05 to 1.45

Below 1.05 the burp gets smaller and cleaner, not longer-awaited. EXACT The wait saturates as the knob approaches 1 (the counted lottery gives about 8,300 sweeps at 1.05 and about 14,000 at 1.001), while the release shrinks to nothing at exactly 1, where curled and flat tie and the tube is no longer above anything. Below 1 the ladder flips. MEASURED, EXPLORATORY In a rehearsal the author calls the backwards reaction, a perfect flat sheet at $\lambda < 1$ sat at zero for hundreds to thousands of sweeps and then curled, one direction first, releasing energy in steps that landed on the exact rungs of the ladder. Above 1.45, MEASURED at 1.5 the tube falls apart within 50 sweeps in every run; EXACT at 1.6 route B costs nothing and at 18 dots no stuck state above flat exists; by a knob setting of 5 to 10 the toy behaves as one with a third square simply forbidden. Nothing curled is stable for now out there.

The warmth g: 1.4 to 2.5, with the maps at 1.5

EXACT Colder, the wait grows like $e^{12/g}$: about 50,000 sweeps at $g = 1$ and nearly three million at $g = 0.75$. MEASURED, EXPLORATORY At $g = 1$, two of four tubes were still tubes after 30,000 sweeps. A laptop cannot wait long enough below about 1, which is why the knob stopped there. Hotter, the tube opens almost at once and from many places, and near $g \approx 3.5$ the flat sheet itself melts into the random phase; that melting point is the one the reservoir estimate of Chapter 7 used. How that melting point moves with the size of the sheet is taken up in a later paper; until then every warm result carries a size caveat.

The size N: 48 to 288 dots

The upper limit was run time on one laptop and the choice to spend compute on repeats rather than size. EXACT The exit count was certified at 48 through 288; the control at 32 dots breaks the pattern, and 16 dots is the knot. MEASURED The front's speed fell roughly as the size to the power -0.7, so longer tubes take longer to finish. OUR READING, UNVERIFIED Whether the wait itself grows, stays flat or falls with size depends on the clock the toy does not fix (Chapter 5). The technical paper claims nothing beyond 288. Networks into the thousands of dots are the subject of a later paper on the scrap.

Stores C: one to two per dot

MEASURED A single store melts every tube; two per dot leaves clean sheets with one scrap. Colder boxes than two per dot were not tried. The store balances were not saved, so they cannot be turned into a calibrated temperature after the fact.

The push s: 4 to 16, then 12 to 160

EXACT Below 12 nothing can leave by single swaps, at the certified sizes; a tube with nothing to spare waits for ever, which is now a permanent test in the code. MEASURED, EXPLORATORY Above, pushes of 12 to 160 units at 64 dots all left, and the hotter the box the more disordered the sheet it ended with, with the ending predicted from the toy's equilibrium curve before the run: 0.97 squares per dot measured against 0.95 predicted at the smallest push, 0.81 against 0.81 at the largest. Give it far more and it melts.

Named or interchangeable dots

OUR READING, UNVERIFIED Every run treats dot 17 as a different thing from dot 41. Giving each distinct wiring one statistical weight, regardless of names, would be a different ensemble; quantum indistinguishability does not by itself require that choice. Energies and the certified 12-unit escape bound stay the same. For the counted A exits at 64 and 96 dots, the corresponding acceptance is reduced by a factor between $N/2$ and N. That does not determine the whole waiting time: other exits and the time spent in neutral arrangements also matter. This paper does not simulate that alternative.

Directions: two

For ideal product grids, four links per dot give two directions; six and eight give three and four. Degree alone does not assign a dimension to an arbitrary network. These larger-degree models (Figure 12) approach the three-direction question and are the subject of a later paper. The tease: a curled direction in three dimensions does not open the way the tube does, and the reason is partly exact and partly still open.

The route

The published route starts from a random tangle; this paper starts from an ordered tube. MEASURED Both were run. From randomness at $\lambda \geq 1$: one hump, no sign of a release at these sizes. From the tube at $\lambda > 1$: a hill, a push, a burp. The two routes are different questions, and the toy gives them different answers.

The point to keep: The box is small and its edges are known. Past most of them lies either an exact statement (nothing below 12 can leave; nothing is below flat), a later measurement with its own paper, or a plain "unknown." The one edge the toy cannot push on its own is the clock.

11 WHAT WE HAVE LEARNED, AND WHAT COMES NEXT

A mechanism we can price, and two questions it opens

Return one last time to the 96-dot tube. Its excess energy is exact: 96 units. The cheapest way out is exact: 12 units at $\lambda = 1.25$, the same at every tube size certified. The counted lottery predicts when it first leaves, and the tube keeps that appointment over a ninety-fold range of waits. Halfway through, 99 dots in 100 belong to one arrangement or the other. When it finishes flat, it gives off exactly one unit per dot, and sealed, the energy is followed to the unit into stores and into a four-dot scrap. These statements come from different tests and parameter ranges; the sealed result is at $\lambda = 1.25$. The fourth bath map passes its criteria from 1.05 to 1.30 at the tested sizes.

What the toy shows about this kind of change

OUR READING, UNVERIFIED The toy gives the larger hypothesis a concrete first question: can a prepared curled arrangement wait, escape and release a calculable amount of energy? Under the specified rules, it can. But its measured endpoint also limits the analogy: removing the four-step curl usually leaves a narrow torus, not two large directions. A fixed first-escape cost is not a measured commitment barrier, and a nonzero ideal energy gap is not a thermodynamic phase-transition test.

THE AUTHOR'S CLAIM The author's wider proposal is that an organized precursor could underlie the origin of space. This paper supplies a controlled toy-model test of part of that proposal, not a test against observations of the universe. Its results remain useful to the next model: the local energy can be priced exactly, the first exit can be predicted, and local flatness is insufficient to ensure large-scale opening: by count most flat wrappings are wide, but the runs stop at the nearest flat arrangement, the narrow one, and their own trapped heat does not widen it. The remainder did not grow, a single front was not established, and three registered maps remained inconclusive before a fourth passed its stated criteria.

What another researcher can build on

Some parts are exact or complete finite listings: the ladder, the 12, the three reachable arrangements, the topology of 120 endpoints. Their scope is the stated rules, sizes and the correctness of the code, which is tested against independent calculations. Other parts are measurements with variation between repeated runs, where a hundred dots in one run are not a hundred experiments. Tests written down before the data are marked pre-registered; the Arrhenius scan, the push threshold and the endpoint-energy reading are exploratory and say so. Interpretations are the author's and her assistants' and

have not been checked by a physicist.

Even what to count is a choice. The simulations count named dots. Counting each wiring once, with the names removed, leaves every energy alone and changes the rates, most for the symmetric arrangements that are easiest to calculate with and that nothing is ever in. That knob is decided and not yet turned.

What the next round of this experiment should record

The fourth map now records the first exit without the old watch, and the saved torus endpoints have been reconstructed. Next, save states after successive swaps and launch repeated continuations from each: how often does each state return to the tube, and how often does it complete conversion? That would distinguish opening the door from committing to the change. Register endpoint topology and loop lengths as outcomes in their own right. Record each store's balance and test equilibration before inferring temperature. A clock with a specified attempt rate per neighborhood would make cross-size kinetic comparisons clearer. Continue predicting a held-out size before reading it.

What comes next

The question that began the work survives its first test. Can an organized arrangement give way sharply, release energy and leave a remainder? In this small laboratory it can, and the first step can be priced to the unit. Three things were left standing at the edge of what this paper measured, and each has work of its own coming, with its predictions written down first.

The scrap. A scrap is the only thing in the new space that still carries the old arrangement, so the next paper, *What the burp leaves behind*, asks what it takes to keep one. MEASURED One result: the scrap counted here is not permanent. Cool the new sheet slowly enough and it heals away, while another kind of leftover survives every cooling we could run.

Three directions. The tube has one open direction, and space has three, so the paper after that, *How curled directions open*, runs the same rules with six links per dot. MEASURED One result: escape from the four-link tube does not settle the larger-direction question. Later runs find a separate barrier for each curled direction. Some reach broader shell profiles, but that does not establish a fixed-budget route that opens every direction.

A rule that pays for width. The toy's rule charges only for loops of four, which is why the first re-gluing is paid for and the next is not. The author's next test adds a small price on loops of six, the same rule one loop-length up, so that a torus six around is itself stable for now and has somewhere to go. That test has now been run at two sizes, with a third under way. Its results belong to a later paper and are not reported here.

Twelve units start an escape. What opens a space?

Continuing into the technical paper

This edition follows the technical paper [9], which holds the equations, tables, uncertainty calculations and full references. Every number above is taken from it or from the project's record, and the kinds-of-statement tags follow the project's own rule that no statement is promoted from one kind to another without a dated note saying why. Its sections run in the same order as the chapters here: the toy and its exact results (Chapters 2 and 3), the cost of leaving (4), the first exit (5), the conversion (6), the sealed system (7 and 8) and the dependence on λ (9).

A few translations for the next step

In the paper	What to keep in mind
"The model"	In the technical paper, our toy: the published model with the knob λ. Only $\lambda = 1$ is the published model, combinatorial quantum gravity.
N	Number of dots. The running example uses $N = 96$.
Torus	Any grid that wraps both ways. The flat torus is the sheet, the curled torus is the tube, and the smallest one, curled both ways, is the knot.
$4 \times L$	Four steps around the short wrap and L around the other, L even and above 4. 4×4 is the knot.
λ **and** g	Lambda sets the price of a surplus square and so the whole ladder; g is the bath's warmth, which sets how often an uphill swap is accepted.
$H = 16(N - S) + 4\lambda X$	The price list: S squares, X surplus squares. Zero on the flat sheet; $4(\lambda - 1)$ per dot on the tube.
$\Delta E^{\ddagger} = 32 - 16\lambda$	The rim: the cost of route A, 12 at $\lambda = 1.25$. Route B costs $64 - 40\lambda$.
d	The square-count signature: pairs of links at a dot that close no square. Sheet 2, tube 1, knot 0. Not a certified geometry.
f	The progress meter, 0 for the tube and 1 for the sheet. Disorder raises it too.
Arrhenius law	Uphill events becoming much rarer as the bath cools, as $e^{-\mathrm{cost}/g}$. Here its two ingredients are counted, not fitted.
Memoryless	The chance of leaving is the same on every attempt, like rolling a die until a six comes up; a long wait so far makes leaving no nearer.
Metastable	Stable for now: escaping the tube's neutral family costs energy. The map's edge near 1.37 is where the wait drops below the watch, not where that stops being true.
Microcanonical	Sealed: total energy fixed. Inferring temperature from the stores also requires a checked equilibrium distribution.
Pre-registered	The prediction and the rules of the test were committed before the run. Exploratory runs say so.
95% interval	A range from a statistical procedure that, under its assumptions, covers the truth 95 times in 100.

Where the evidence is recorded

The technical manuscript and its methods supplement are in the recorded repository version, and the supplement maps every claim to code, configurations and result files. Experiment labels in that record: T7 and T8 (the decay and the map), T9 (the sealed reservoir), T10 and T11 (the scrap and its position), T22 (exits counted move by move), T38 (the long-wait test). The questions teased above, what the scrap does afterwards, larger networks and more directions, have their own labels in the record and their own papers. The push threshold covers 48 to 192 dots and the exit enumeration 48 to 288; neither extends to other sizes.

Disclosure

The author used Claude (Anthropic) and ChatGPT/Codex (OpenAI) to assist with code, analysis and writing, and is responsible for the result. This edition was drafted with Claude from the corrected manuscript and the project's record. Its analogies and pictures introduce no new experiments or evidence. No physicist has reviewed any of it.

Sources

[1] C. A. Trugenberger, *Combinatorial quantum gravity: geometry from random bits*, J. High Energy Phys. **09** (2017) 045, arXiv:1610.05934. Cited as in the technical paper.

[2] C. Kelly, C. A. Trugenberger, and F. Biancalana, *Self-assembly of geometric space from random graphs*, Class. Quantum Grav. **36**, 125012 (2019), arXiv:1901.09870.

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[4] A. Gorsky and O. Valba, *Interacting thermofield doubles and critical behavior in random regular graphs*, Phys. Rev. D **103**, 106013 (2021), arXiv:2101.04072.

[5] C. Kelly, *Emergent geometry and discrete curvature in random graphs*, Ph.D. thesis, Heriot-Watt University (2022), `https://www.ros.hw.ac.uk/handle/10399/4591`.

[6] C. A. Trugenberger, *Dark matter and dark energy in combinatorial quantum gravity*, arXiv:2409.09385 (2024).

[7] M. Creutz, *Microcanonical Monte Carlo simulation*, Phys. Rev. Lett. **50**, 1411 (1983). Cited as in the technical paper; we have not read it in full.

[8] S. Coleman, *Fate of the false vacuum: semiclassical theory*, Phys. Rev. D **15**, 2929 (1977). Described here from general knowledge of the paper; we have not read it in full.

[9] E. Smith, *The curled torus burps: activated escape and conversion in a graph model of emergent geometry*, technical manuscript, 11 October 2026, in the project repository.

[10] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[11] The written search behind this sentence, 10 October 2026: where we looked, with what terms, and what came closest. File `docs/reading/notes/2026-10-10_prior_work_lambda_above_one.md` in the project repository.

[12] C. Kelly and C. A. Trugenberger, *Combinatorial quantum gravity: emergence of geometric space from random graphs*, arXiv:1811.12905 (2018).

Sources 1 to 8 carry the same numbers as in the technical paper [9]; 9 to 12 are this edition's own.